



Foundation models in medical imaging

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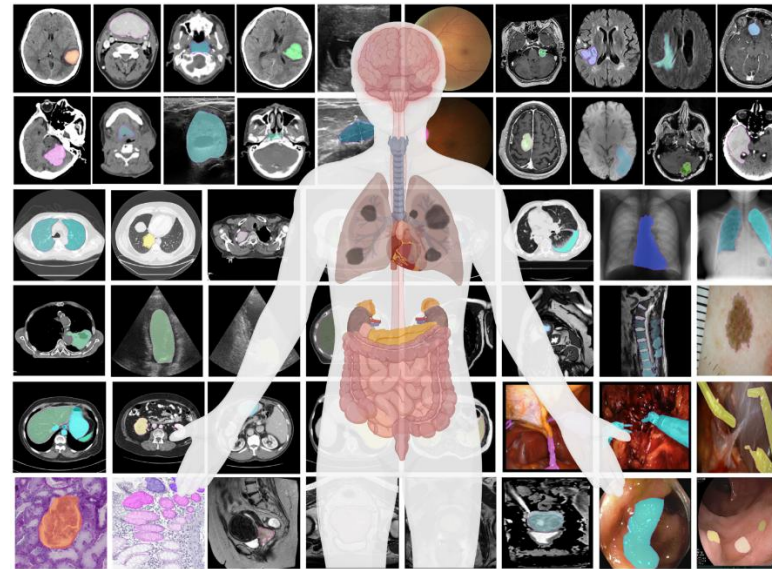
Motivations - Biomedical imaging

High/multi dimensionality

- 2D vs 3D
- 1024x1024 px
- 256x256x256 px

Multimodality

- MR / CT / US / Optics
- EEG / ECG / Genetics
- Report



Annotations

- Interobserver variability
- Quality

Scarcity

- Annotations
- Image accessibility

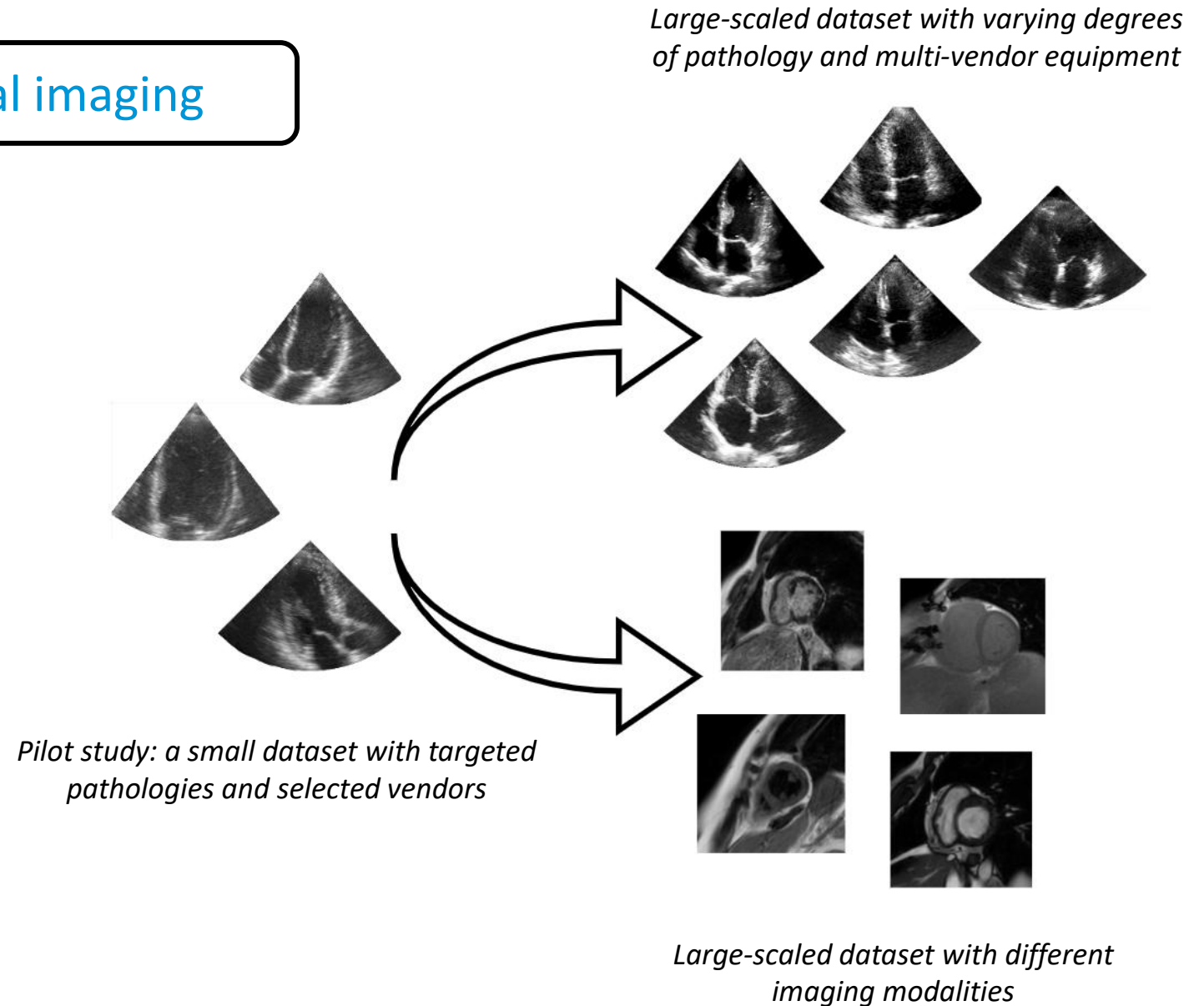
Domain shifts

- Multicentre
- Multivendor
- Acquisition settings

Motivations - Biomedical imaging

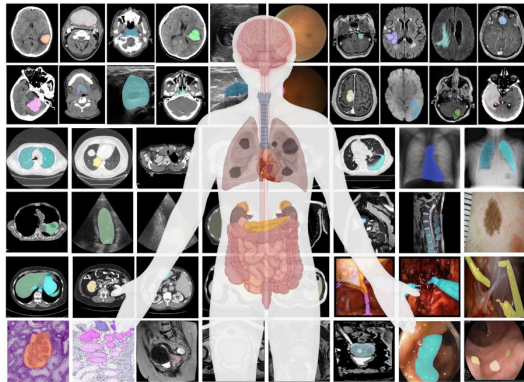
Needs for generalization

- ▶ On large scale dataset
- ▶ Through modalities
- ▶ Through pathologies

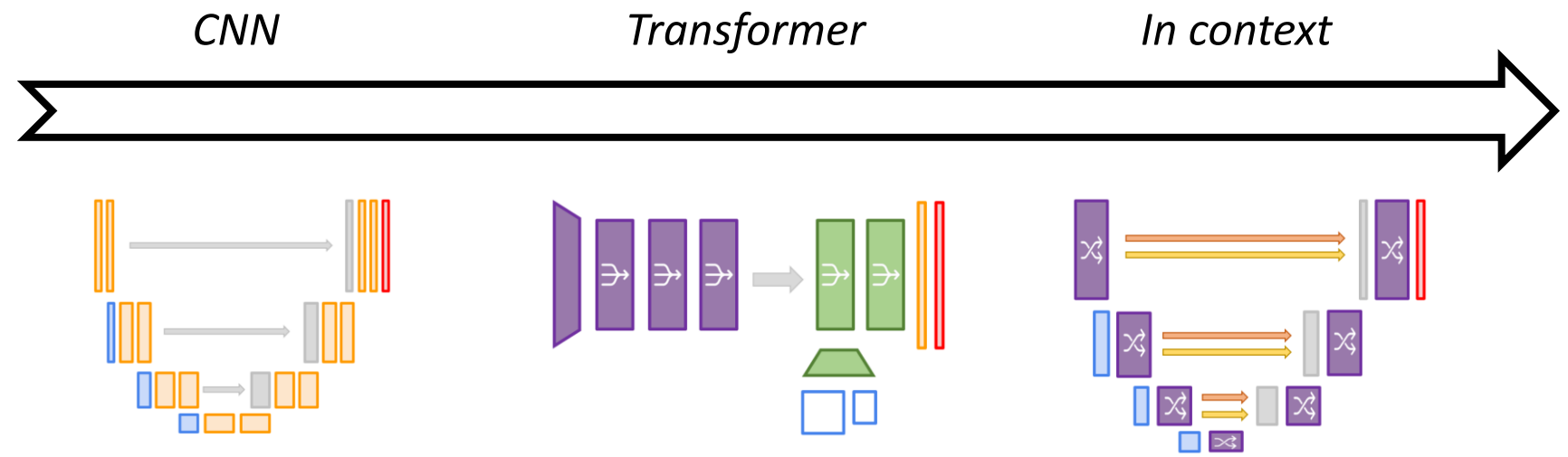


Foundation models - Definition

- ▶ A large-scale pretrained model
- ▶ Adapting to multiple tasks with little or no fine-tuning



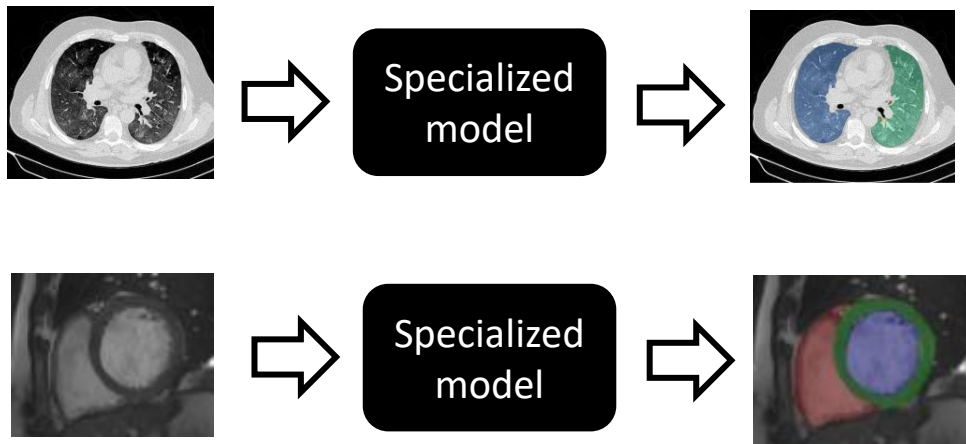
*Large-scale dataset :
millions of images*



Foundation models

Before

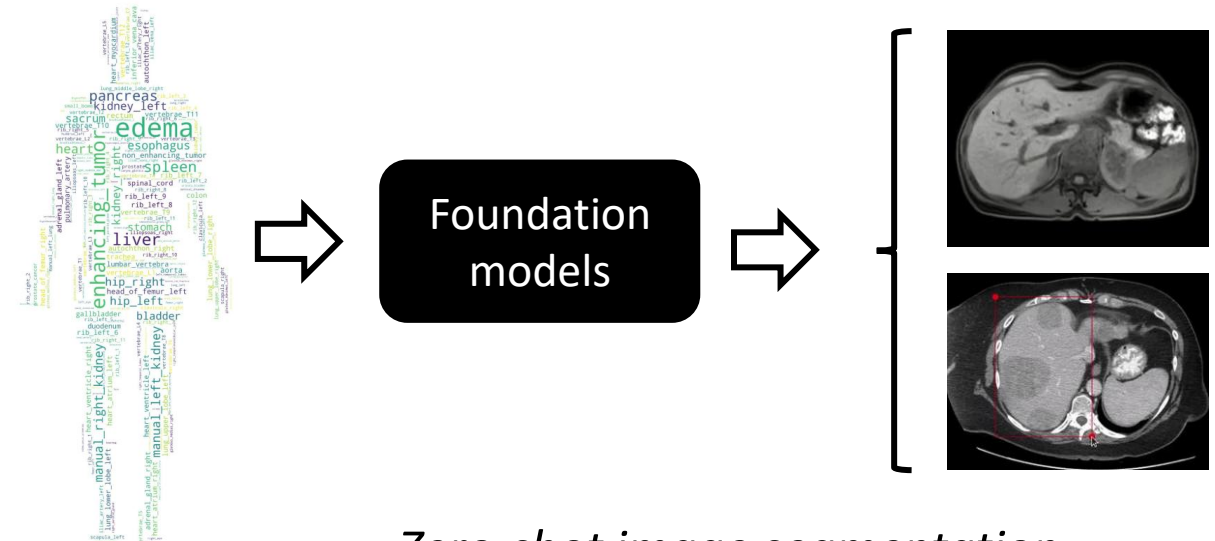
- Specialized model
- Task-specific training
- Annotated dataset



Training / inference scheme

After

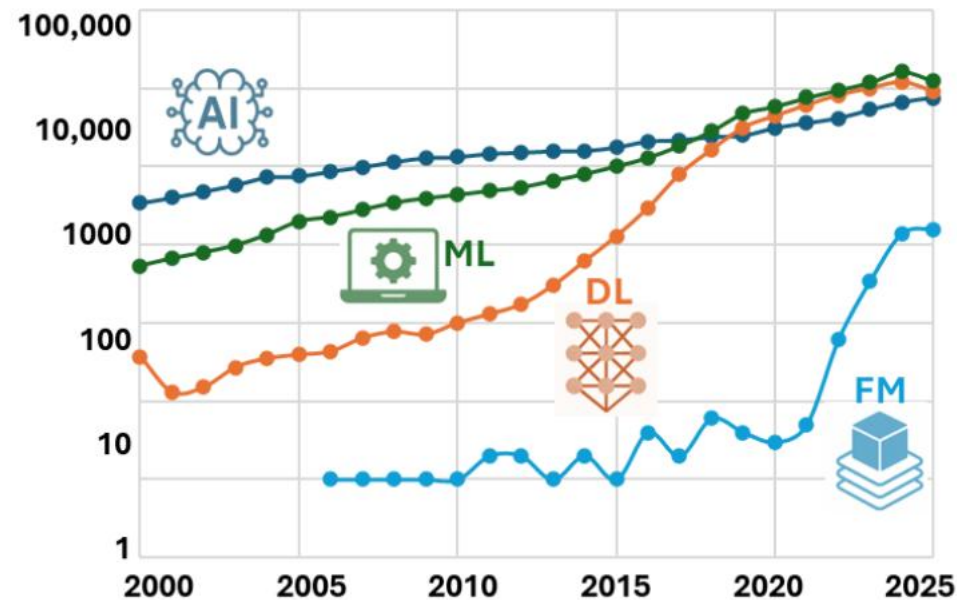
- Generic model
- Zero-shot inference
- Few-shot learning



Zero-shot image segmentation

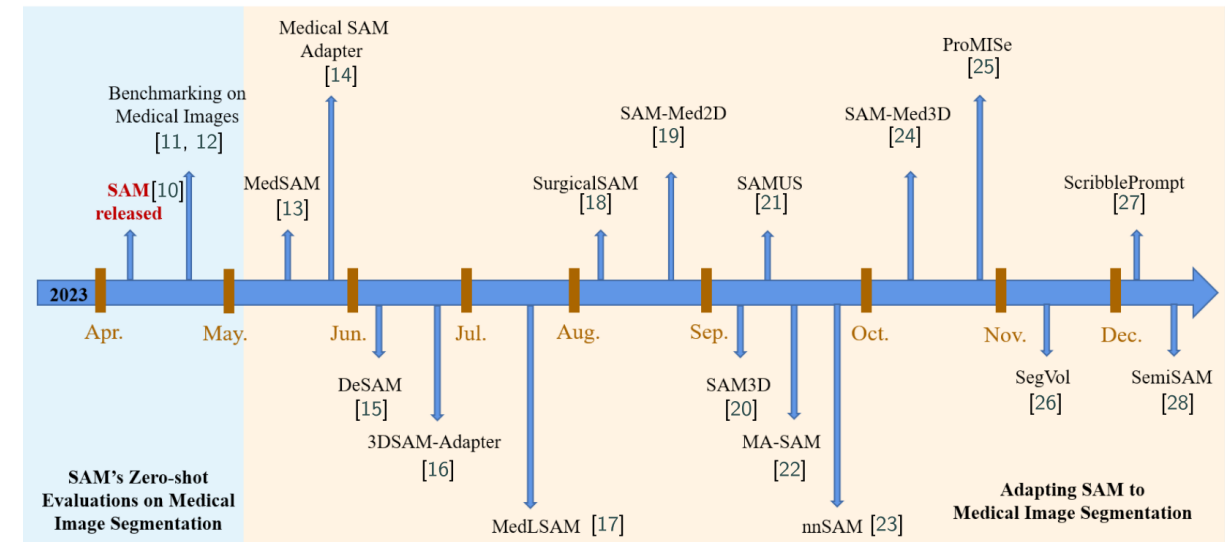
Motivations

Publication trends



Source: Scopus, September 2025

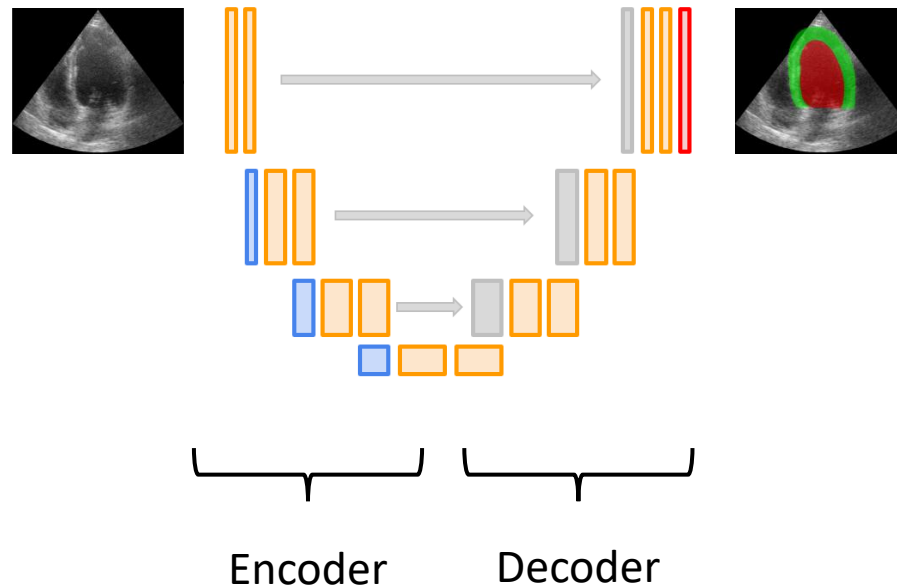
8 months of publications between 2023 and 2024



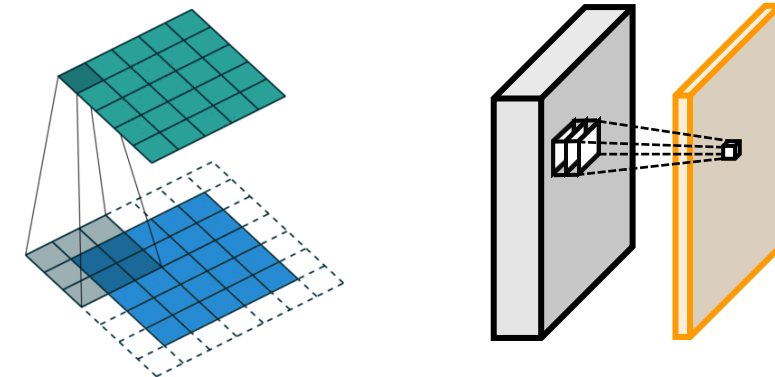
Source: [Zhang et al., CIBM, 2024]

Before foundation models - CNN-based approaches

- Create relevant information called feature map (convolution + non linear function)
- Encoder / decoder architecture



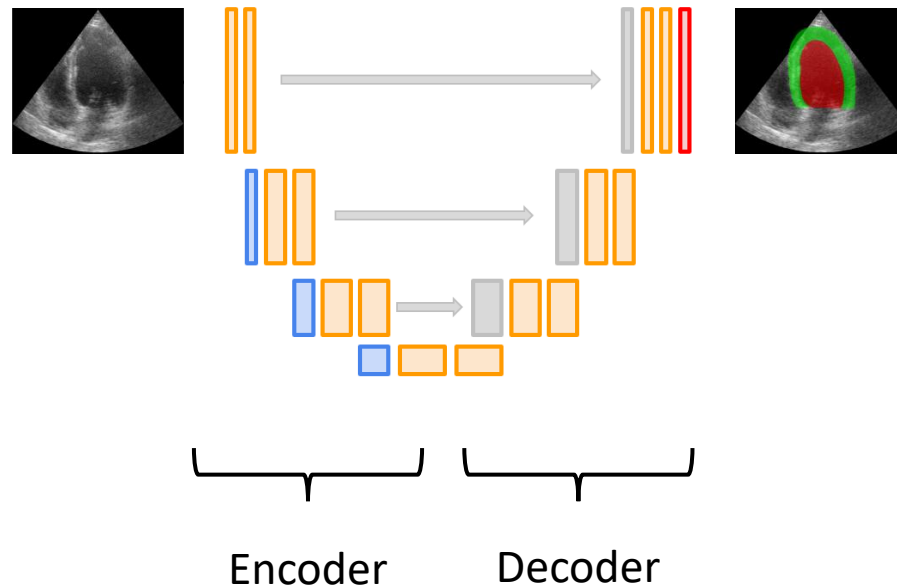
Convolution operation



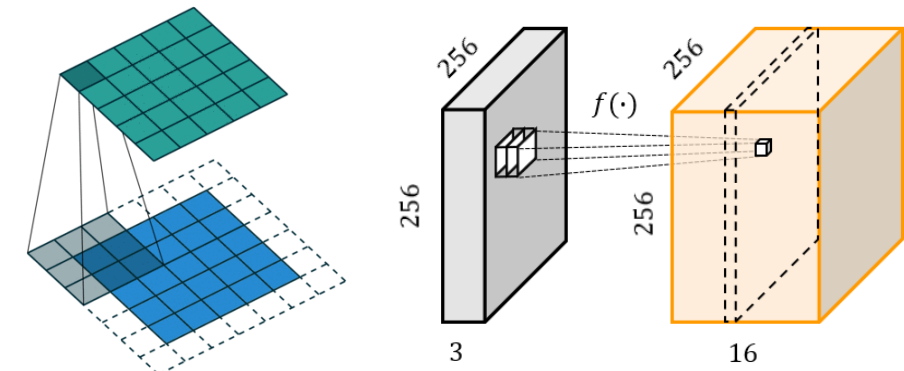
Filter of size 3×3
 $\# param = 3 \times 3 \times 3 + 1 = 28$

Before foundation models - CNN-based approaches

- Create relevant information called feature map (convolution + non linear function)
- Encoder / decoder architecture



Feature maps

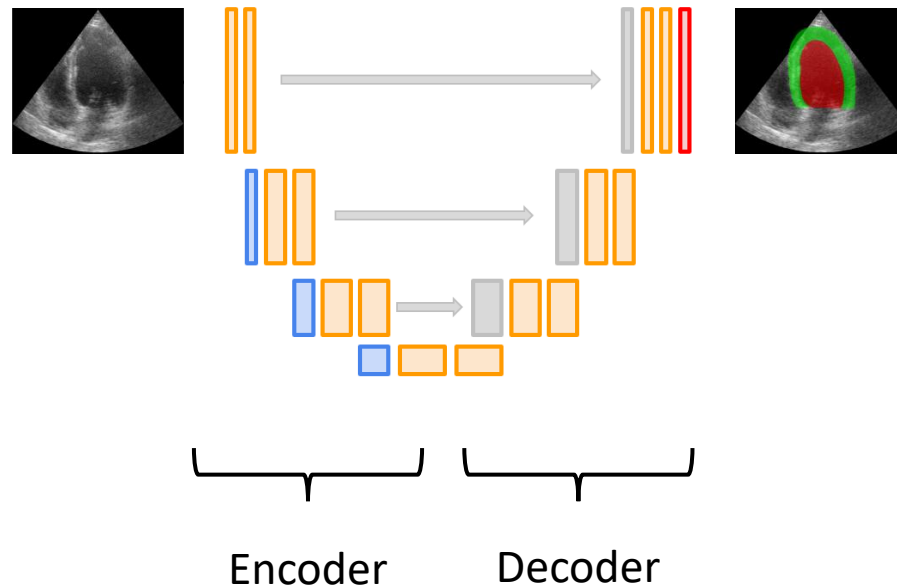


Filter of size 3×3

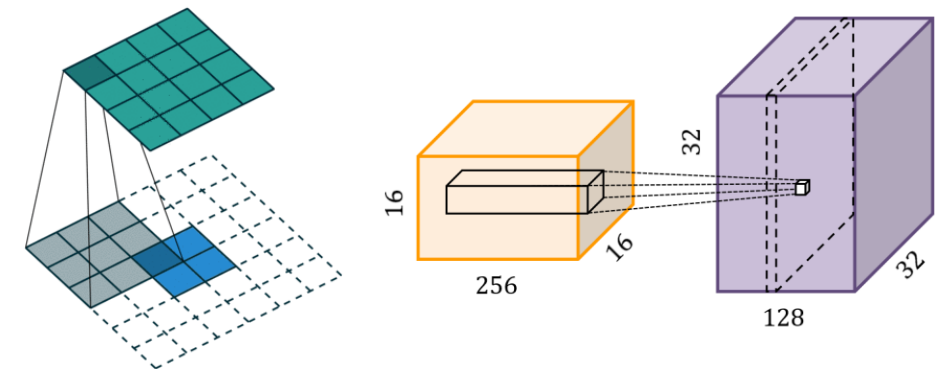
$$\# param = 16 \times (3 \times 3 \times 3 + 1) = 448$$

Before foundation models - CNN-based approaches

- Create relevant information called feature map (convolution + non linear function)
- Encoder / decoder architecture



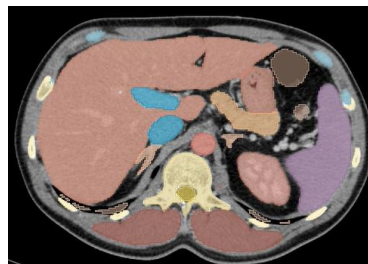
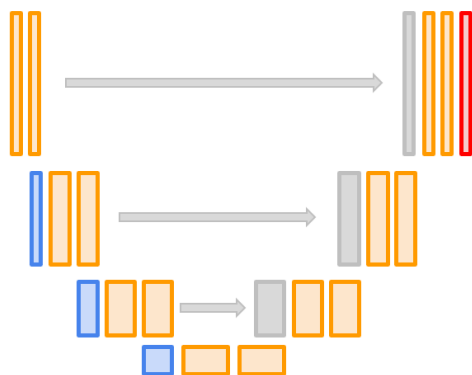
Deconvolution operation



$$\begin{aligned} \text{Filter of size } 3 \times 3 \\ \# \text{ param} &= 128 \times (3 \times 3 \times 256 + 1) \\ &= 295,040 \end{aligned}$$

Before foundation models - TotalSegmentor

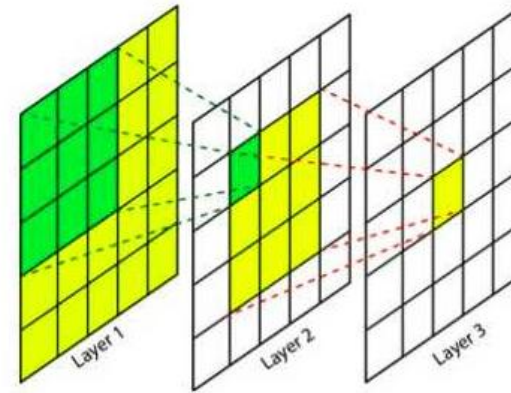
- ▶ 40 M of parameters (depending on the GPU)
- ▶ 3D images from publicly available medical datasets
- ▶ Based on nnU-Net algorithm



- Multimodal model: CT, MRI
- CT model trained on:
 - 1204 CT
 - 104 anatomical structures
 - 27 organs, 59 bones, 10 muscles, 8 vessels
- CT/MRI model trained on:
 - 527 CT and 616 MRI
 - 80 anatomical structures

Limitations of CNN-based approaches

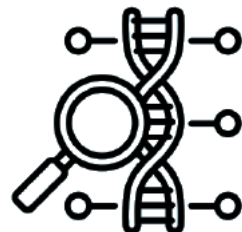
- Receptive field
 - Different parts of an image communicate in deep layers



- How to integrate images with others modalities?



Medical images



Genetics



Reports



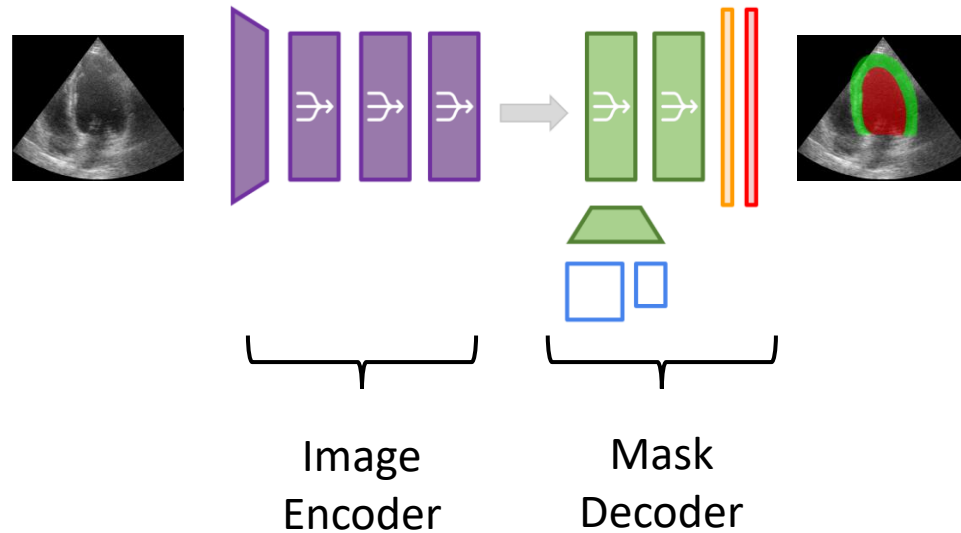
Lab tests



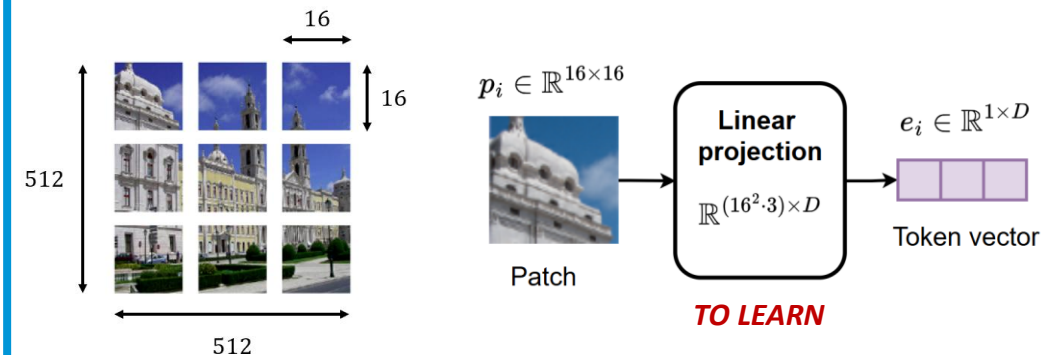
EHR

Foundation models - transformer-based approaches

- Create relevant information called tokens (tokenization + attention + non linear function)
- Encoder / decoder architecture



Tokenization procedure

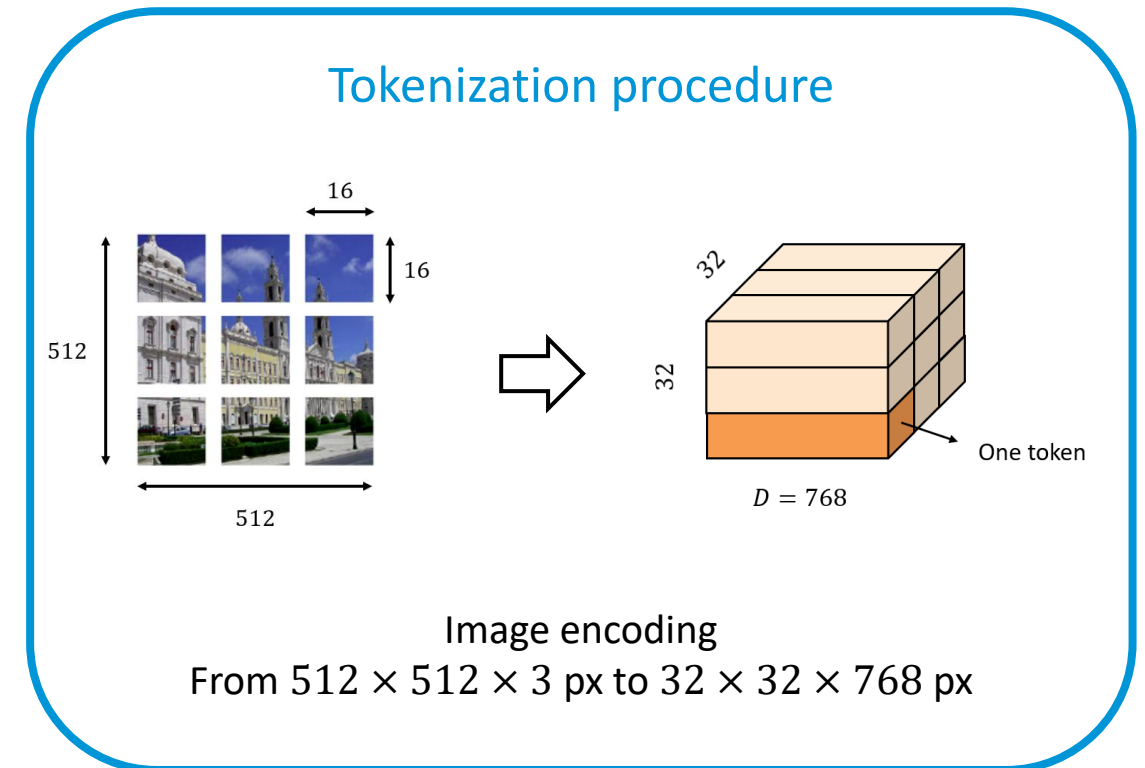
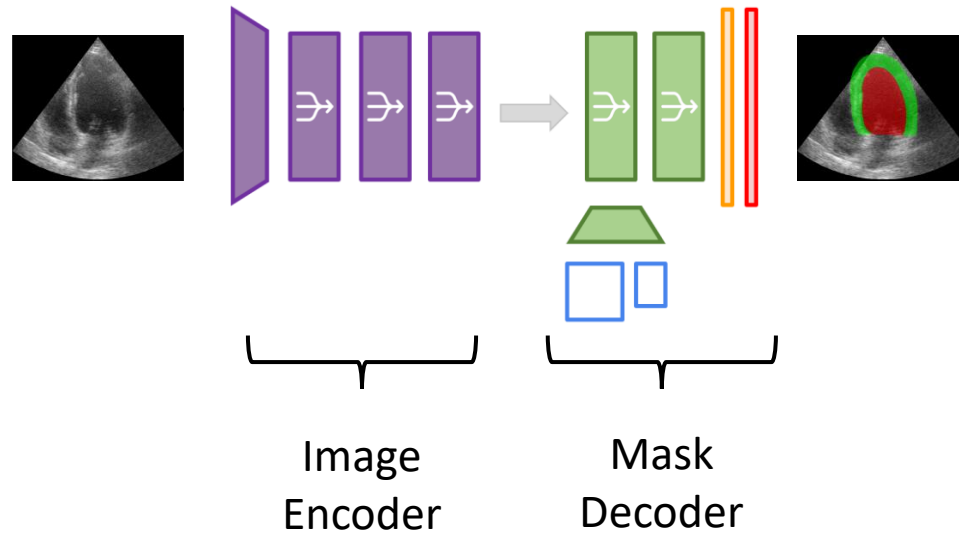


Token size $D = 768$

param = $16 \times 16 \times 3 \times D = 589,824$

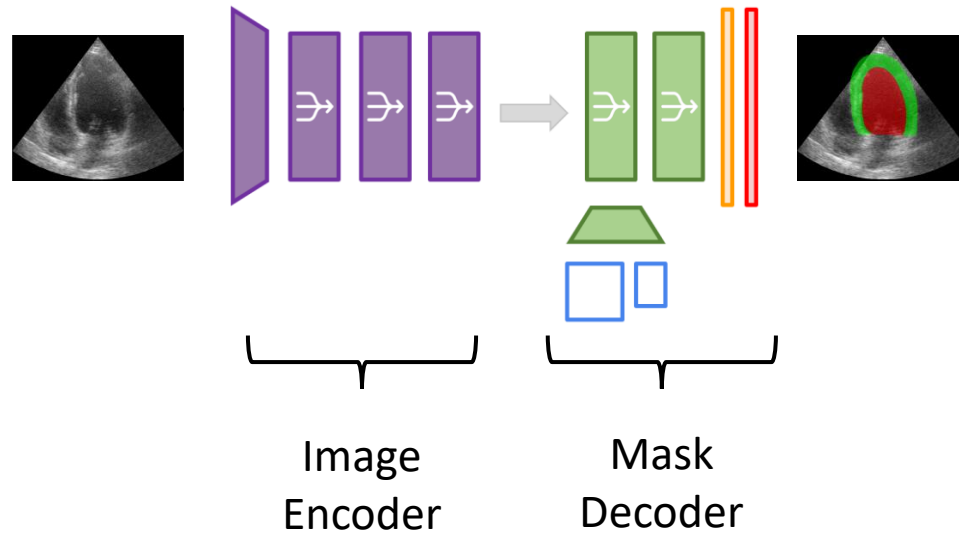
Foundation models - transformer-based approaches

- Create relevant information called tokens (tokenization + attention + non linear function)
- Encoder / decoder architecture

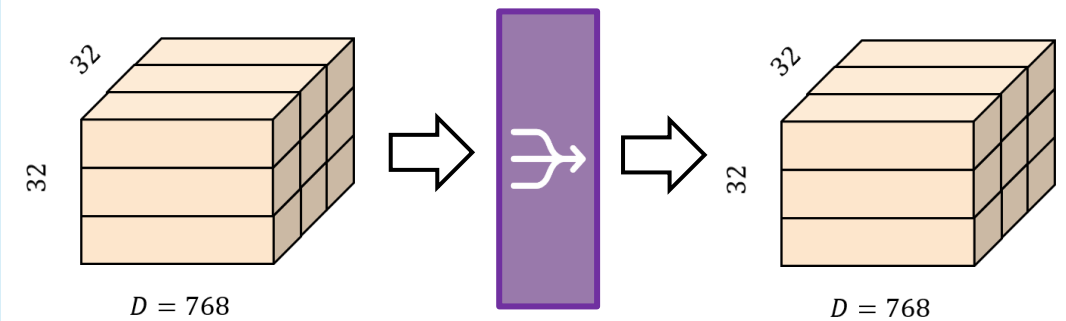


Foundation models - transformer-based approaches

- Create relevant information called tokens (tokenization + attention + non linear function)
- Encoder / decoder architecture



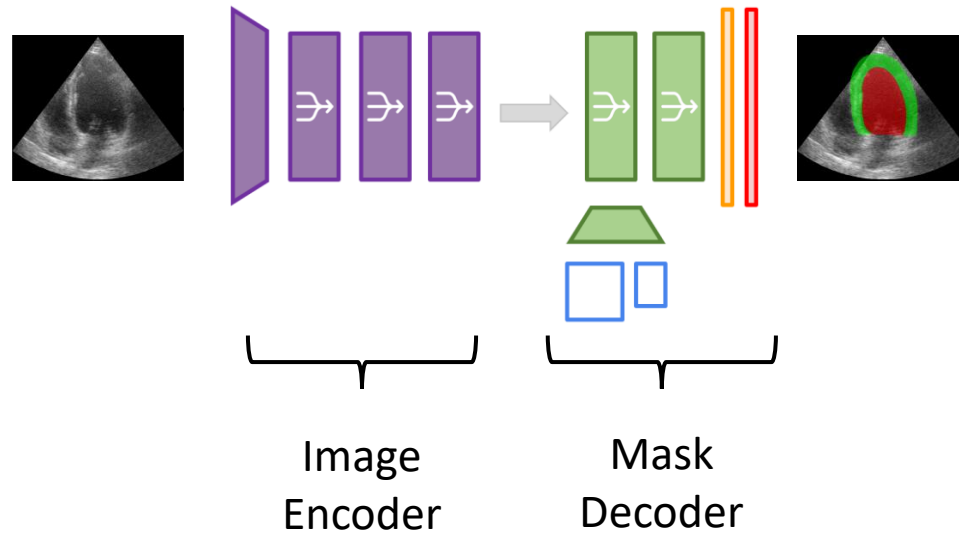
Transformer blocs / layers



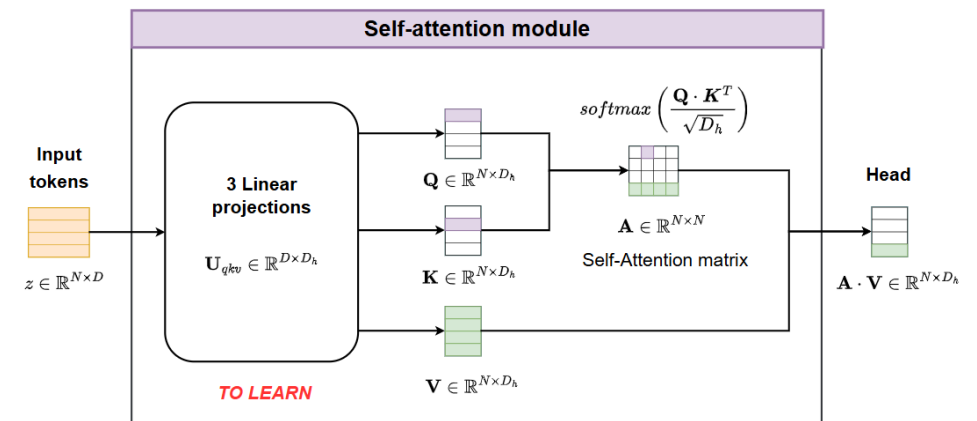
Multi-head attention module + non linearity

Foundation models - transformer-based approaches

- Create relevant information called tokens (tokenization + attention + non linear function)
- Encoder / decoder architecture



Self-attention module

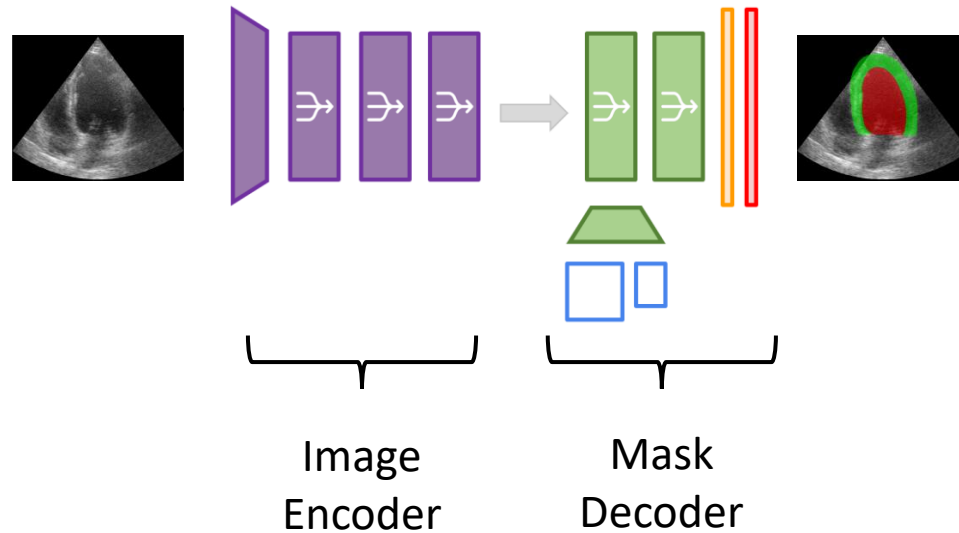


$$D = 768, D_h = 64$$

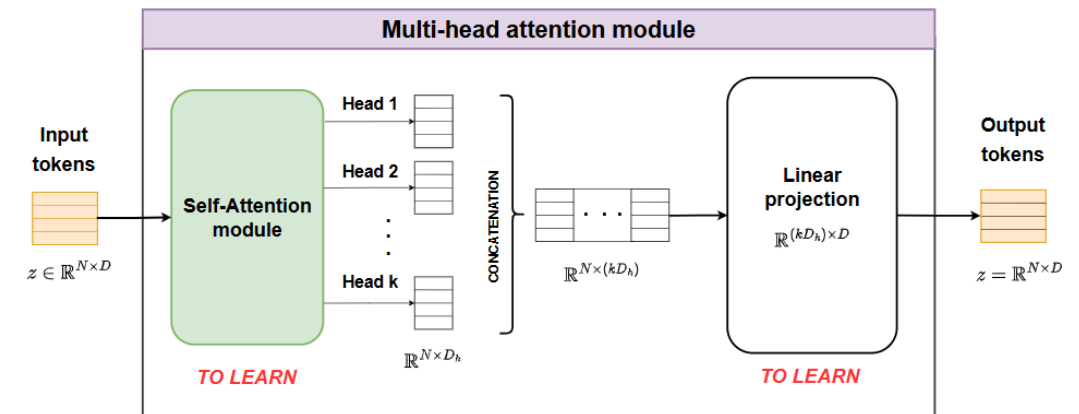
$$\# \text{ param} = 3 \times 768 \times 64 = 147,456$$

Foundation models - transformer-based approaches

- Create relevant information called tokens (tokenization + attention + non linear function)
- Encoder / decoder architecture



Multi-head attention module

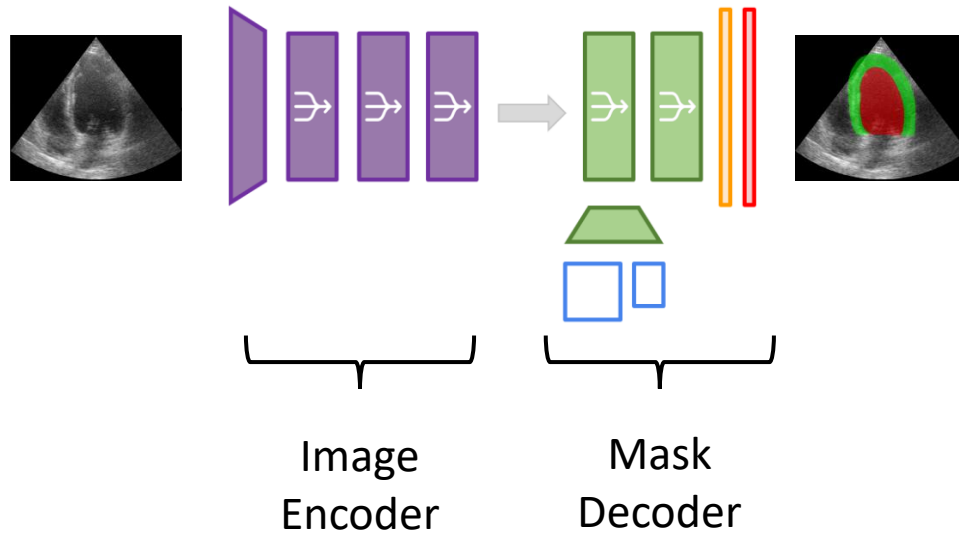


$$D = 768, D_h = 64, k = 12$$

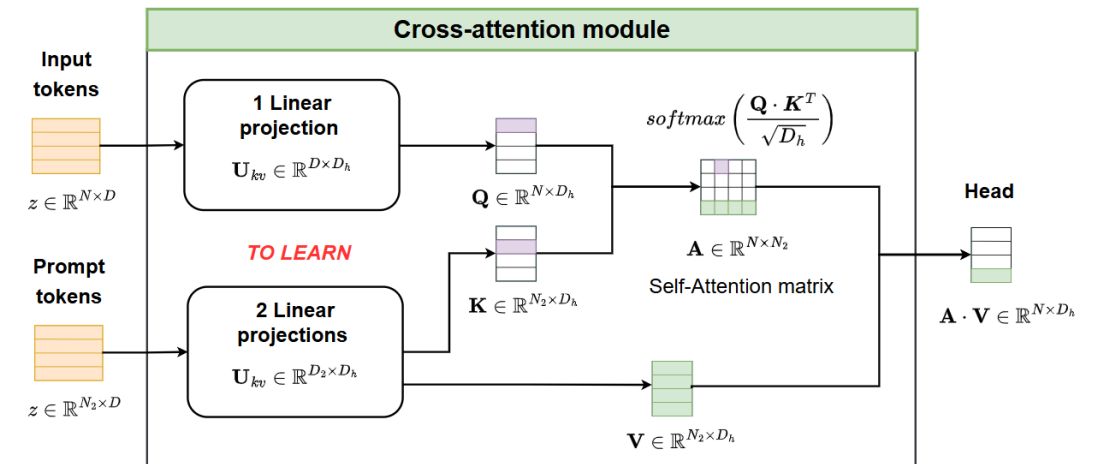
$$\# param = 12 \times 3 \times 768 \times 64 + 768 \times 768 = 2,359,296$$

Foundation models - transformer-based approaches

- Create relevant information called tokens (tokenization + attention + non linear function)
- Encoder / decoder architecture



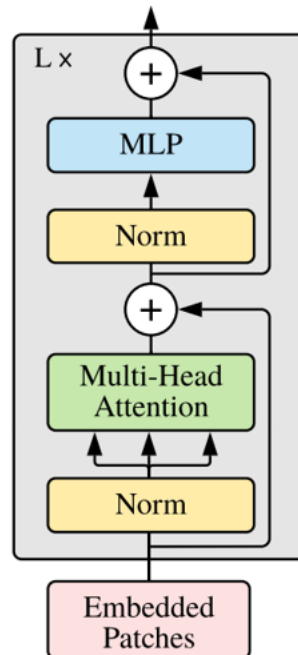
Cross-attention module



Foundation models - transformer-based approaches

- ▶ ViT: Visual Transformer
- ▶ Reference model to encode images

Transformer Encoder

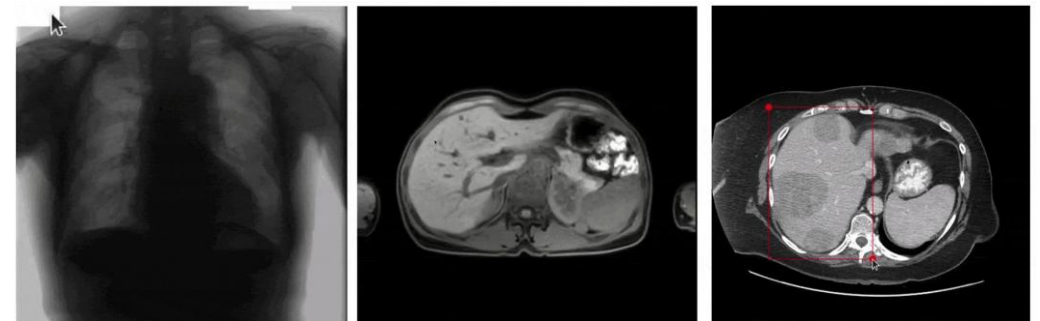
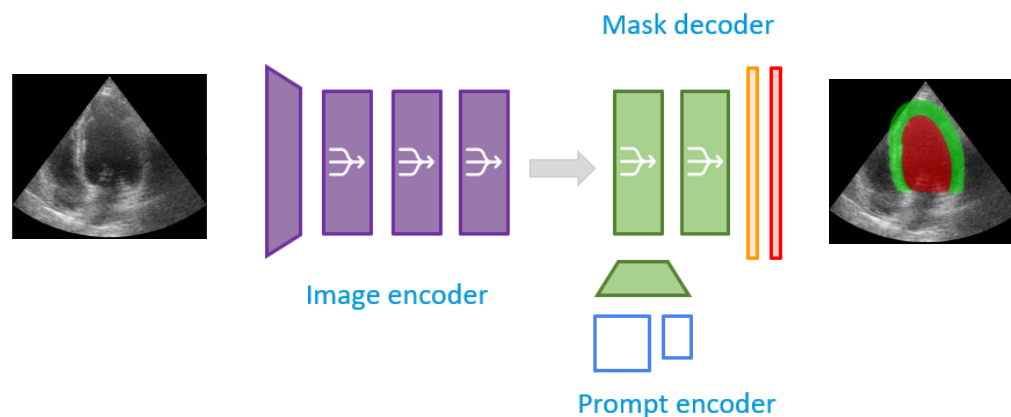


Models	Nb layers	Token size D	Nb heads	Nb parameters
ViT-Base	12	768	12	86 M
ViT-Large	24	1024	16	307 M
ViT-Huge	32	1280	16	632 M

Foundation models - MedSAM

- ▶ 91 M of parameters
- ▶ Image encoder - Vit-B
- ▶ 2D images
- ▶ From publicly available medical datasets

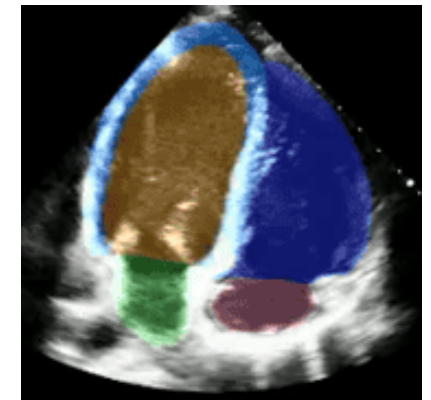
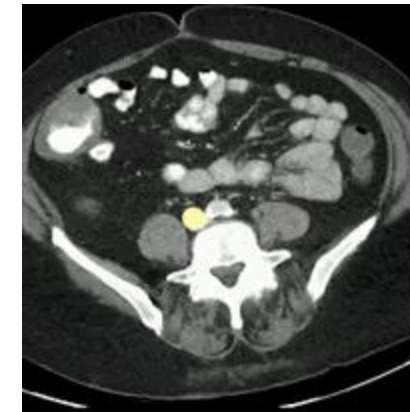
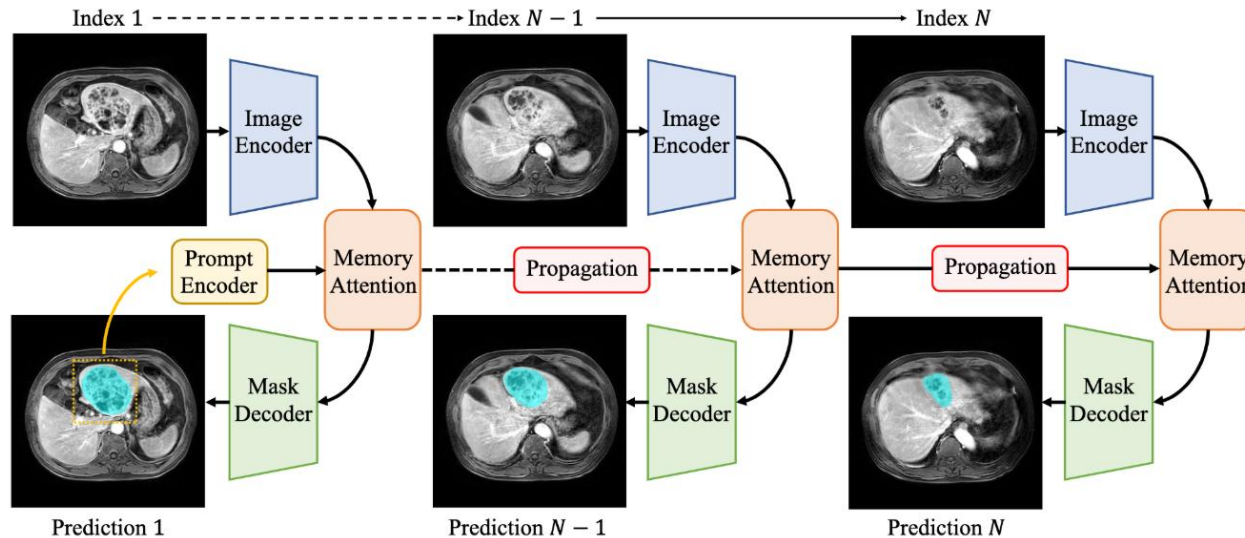
- 1.5 M 2D image-mask pairs
- 10 imaging modalities
- 30 cancer types



Foundation models - MedSAM2

- ▶ 40 M of parameters
- ▶ Image encoder – Hiera (Hierarchical ViT)
- ▶ 3D images and 2D+t sequences
- ▶ From publicly available medical datasets

- 455 K 3D image-mask pairs
- 76 K annotated video frames
- Multimodal: CT, PET, MRI, US, endoscopy



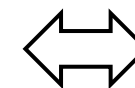
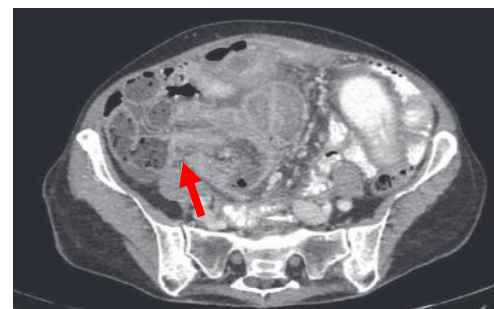
Multimodal foundation models - BiomedClip

- ▶ 200 M of parameters
- ▶ Biomedical vision-language processing
- ▶ Align medical images with text

▶ Targeted downstream tasks:

- Image classification
- Medical visual question answering

- 4.4 M scientific reports from PubMed
- 15 M image-text pairs from fig./caption
- Large diversity of image types

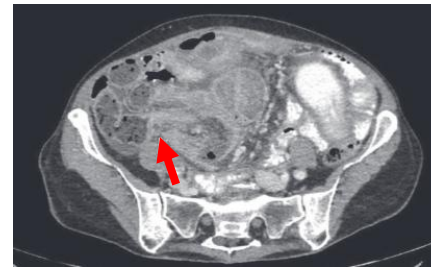


CT images suggestive of intussusception Axial CT image of an apparent small bowel transition point in the right lower quadrant (indicated with red arrow)

Multimodal foundation models - BiomedClip

Image encoder

- ✓ ViT-B/16
- ✓ Patch size: 16x16
- ✓ 86 M parameters

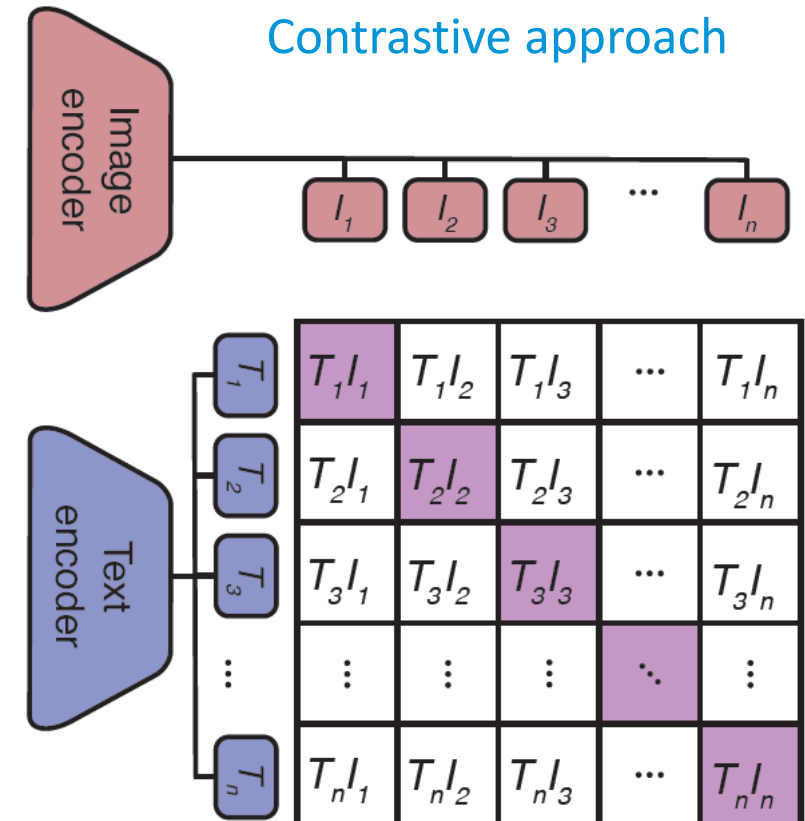


CT images suggestive of intussusception Axial CT image of an apparent small bowel transition point in the right lower quadrant (indicated with red arrow)

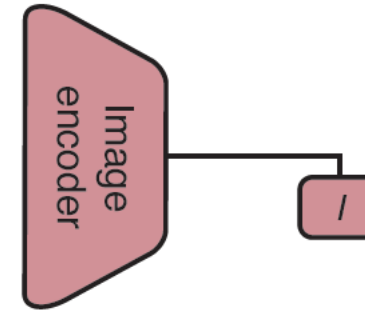
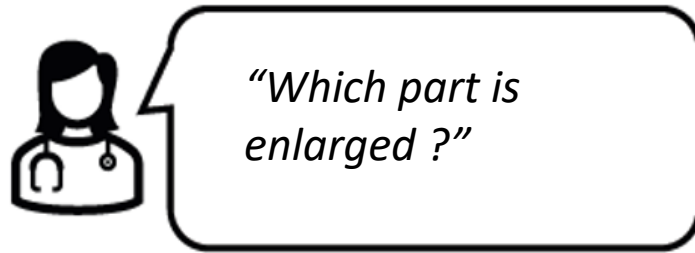
Text encoder

- ✓ PubMedBERT
- ✓ Max. length 256 tokens
- ✓ 110 M parameters

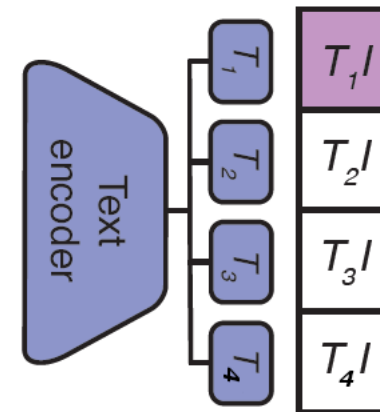
CLIP-based method Contrastive approach



BiomedCLIP - Visual question answering



- T1: The myocardium is enlarged
- T2: This left ventricle is enlarged
- T3: The left atrium is enlarged
- T4: No structure is enlarged

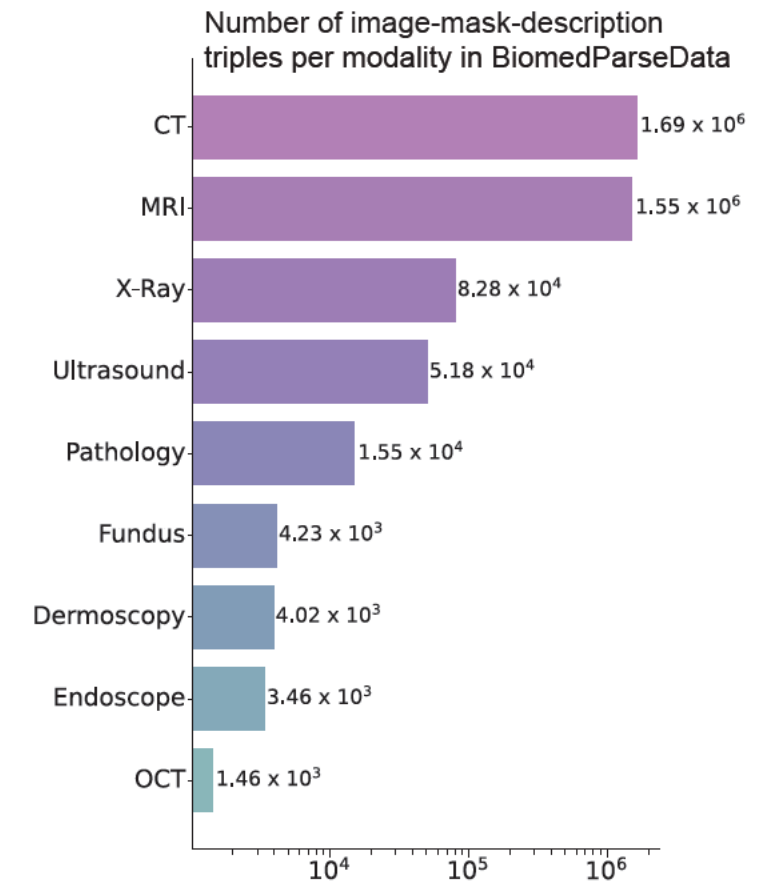


- ☒ The myocardium is enlarged (90 %)
- ☐ The left ventricle is enlarged (2 %)
- ☐ The left atrium is enlarged (3 %)
- ☐ No structure is enlarged (5 %)

Multimodal foundation models - BiomedParse

- ▶ 600 M of parameters
- ▶ Targeted downstream tasks:
 - segmentation / detection / recognition

- 1.1 M images across 9 imaging modalities
- All with manual /semi-manually segmented
- 3.4 M image-masks-label triplets
- 6.8 M image-masks-description triplets
- Across 9 imaging modalities



Multimodal foundation models - BiomedParse

Image encoder: EVA02-CLIP-L-14

- ✓ ViT-L/14 (24 layers, D=1024)
- ✓ 307 M parameters

Text encoder: CLIP text encoder

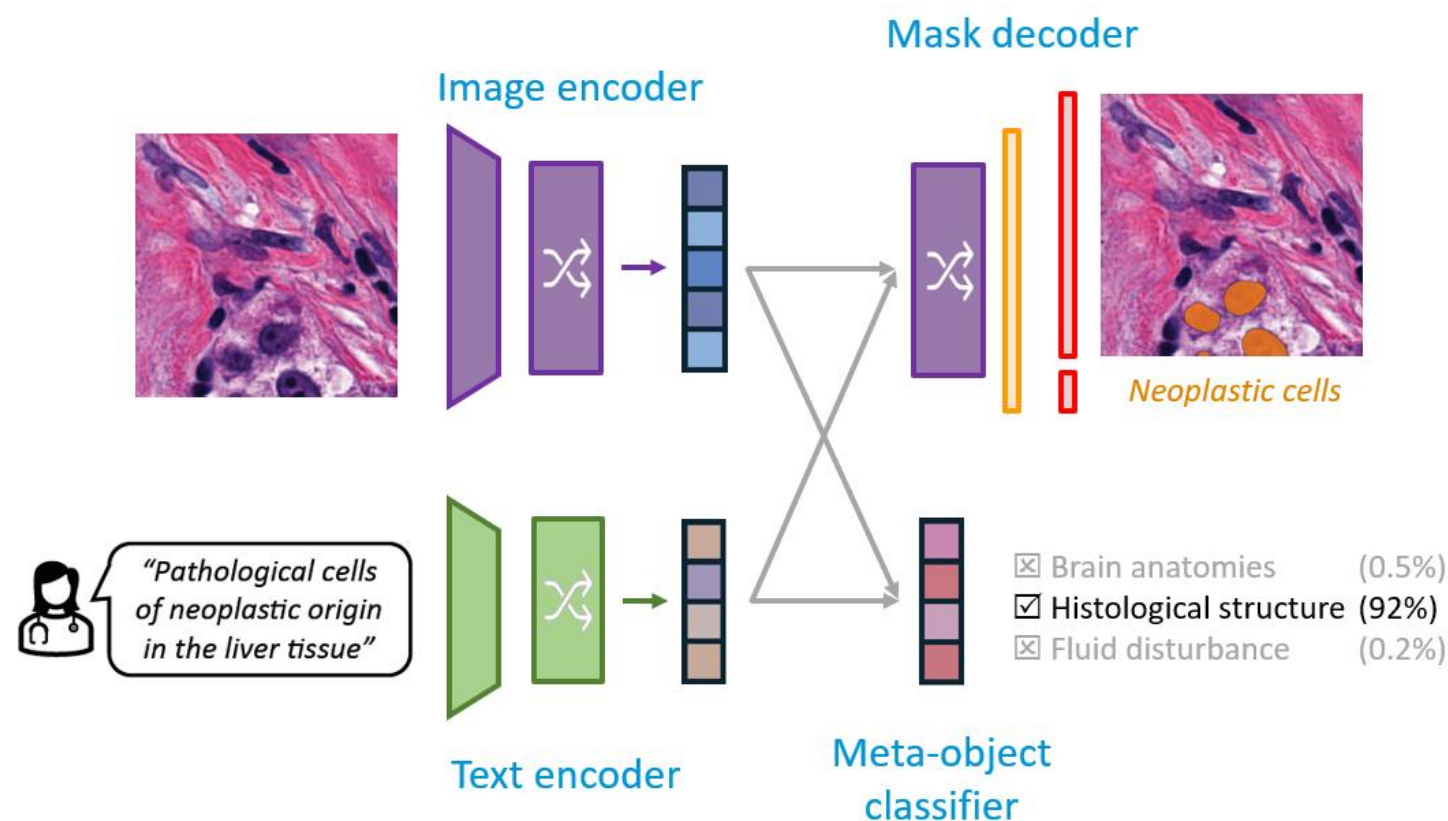
- ✓ Initialized with PubMedBert
- ✓ 124 M parameters

Mask decoder: SEEM

- ✓ 50 M parameters

Meta-object classifier: simple MLP

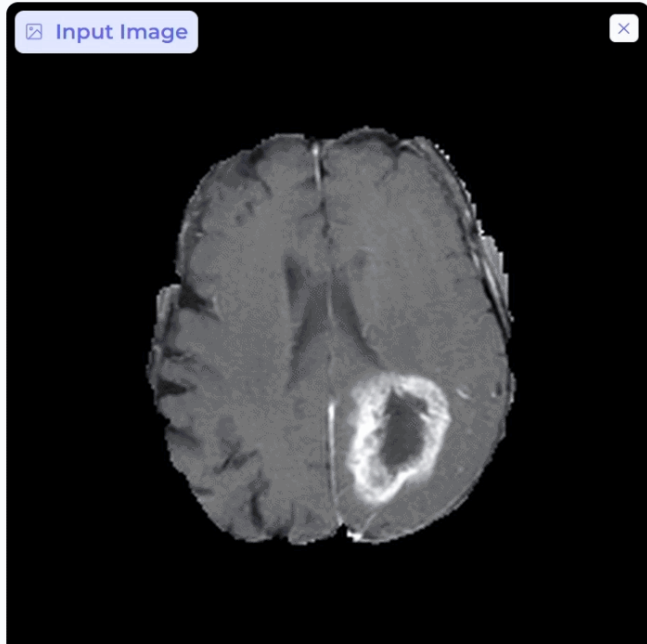
- ✓ 15 meta-objects
- ✓ 300 K parameters



BiomedParse - Segmentation task

From <https://microsoft.github.io/BiomedParse/>

Input Image







Parsing Results



Text prompt

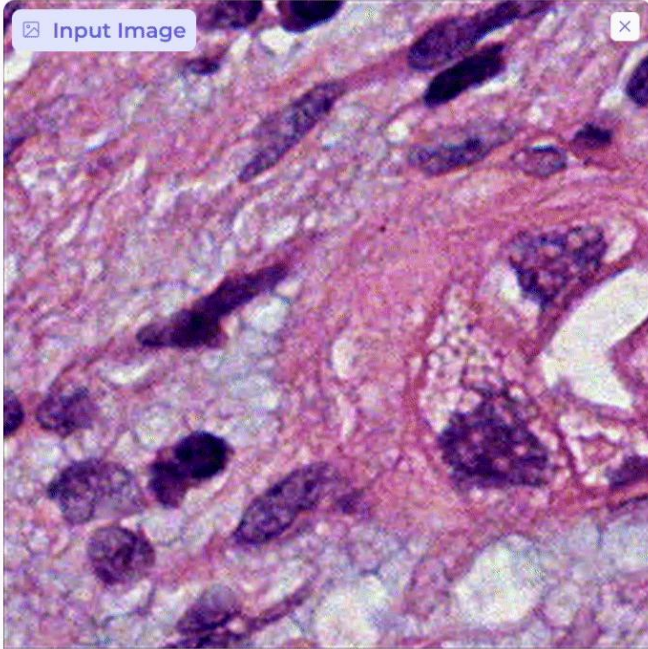
Output text

Go!

BiomedParse - Recognition task

From <https://microsoft.github.io/BiomedParse/>

Input Image







Text prompt

Parsing Results



Output text

Go!

Multimodal foundation models - EchoPrime

- ▶ 200 M of parameters
- ▶ Biomedical vision-language processing
- ▶ Four main steps:
 - Videos / reports alignment
 - View classification
 - Anatomy attention
 - Cross-modal retrieval
- ▶ Targeted task:
 - Automatic report generation

- 108 K patients
- 275 K echocardiographic exams
- 1 exam: >40 videos + 1 clinical report
- 12 M videos
- Structuration of the reports per
anatomical section / **interpretation task**

Left ventricle (LV):

LV ejection fraction is 56%

The LV size is normal

Aorta valve:

A bioprosthetic valve is present

Multimodal foundation models - EchoPrime

Videos / report alignment

Videos encoder

- ✓ mViT-S (16 layers, D=768)
- ✓ 35 M parameters

Text encoder

- ✓ BioMedBERT
- ✓ 110 M parameters

Each videos is aligned with its report independently

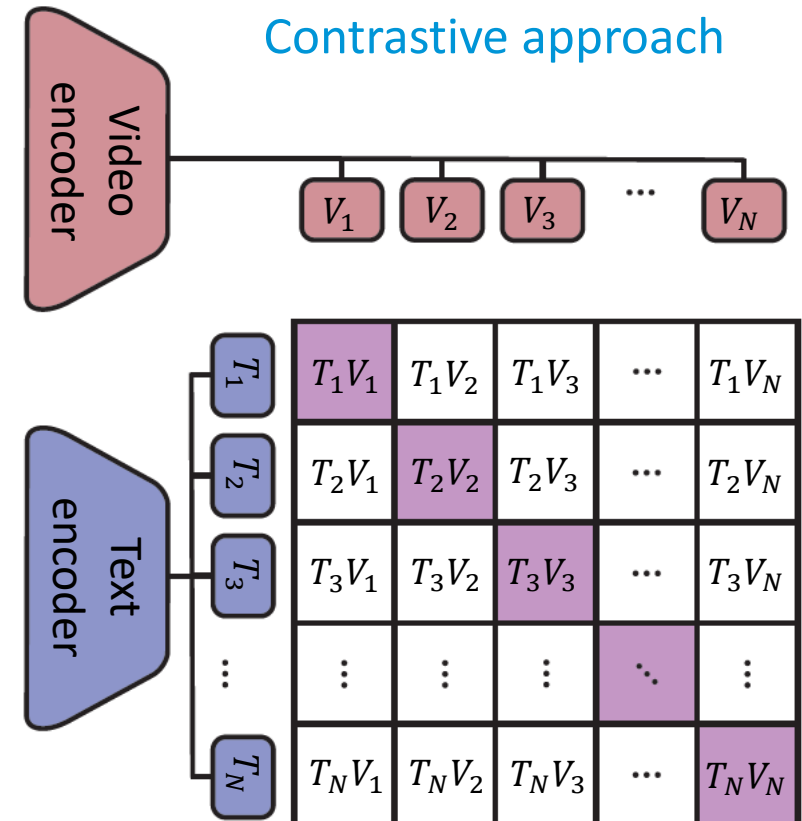


Left ventricle:
LV ejection fraction 60%
Mid diastolic dysfunction

Aortic valve:
A bioprosthetic valve is present

Pericardium:
Small pericardium effusion

CLIP-based method Contrastive approach



Multimodal foundation models - EchoPrime

View classification

Model

- ✓ ConvNextBase
- ✓ 28 M parameters

Pipeline

- ✓ Video as input
- ✓ One-hot vector as output
- ✓ Classify 58 views

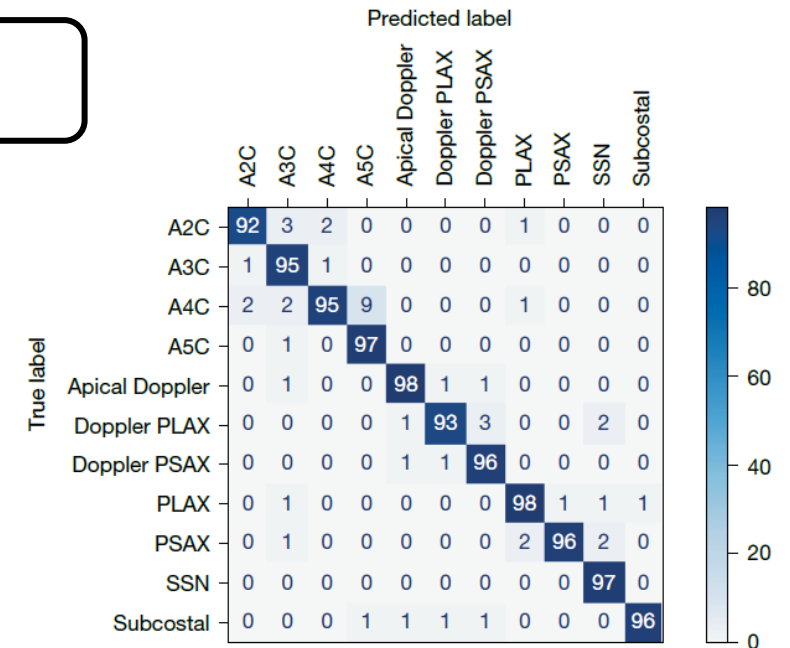


Apical Doppler

Doppler PLAX

A4C

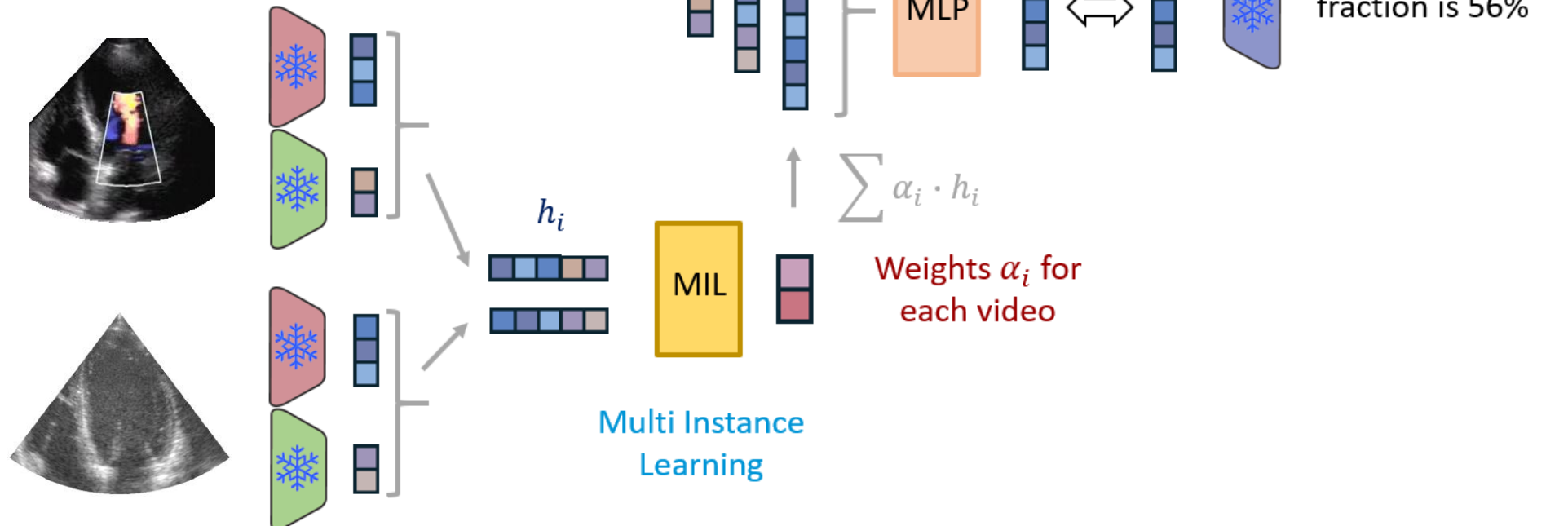
A2C



Multimodal foundation models - EchoPrime

Anatomical attention module

- Prioritization of echocardiographic views for different anatomical structures

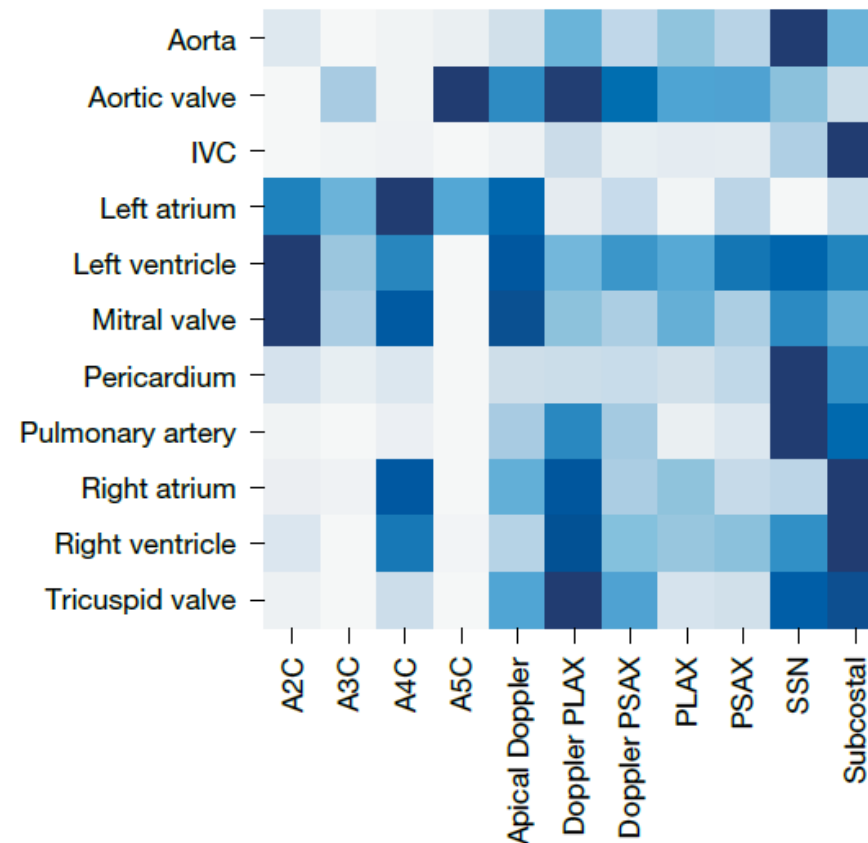


Multimodal foundation models - EchoPrime

Anatomical attention module

- Training procedure repeated for each interpretation task within each anatomical section

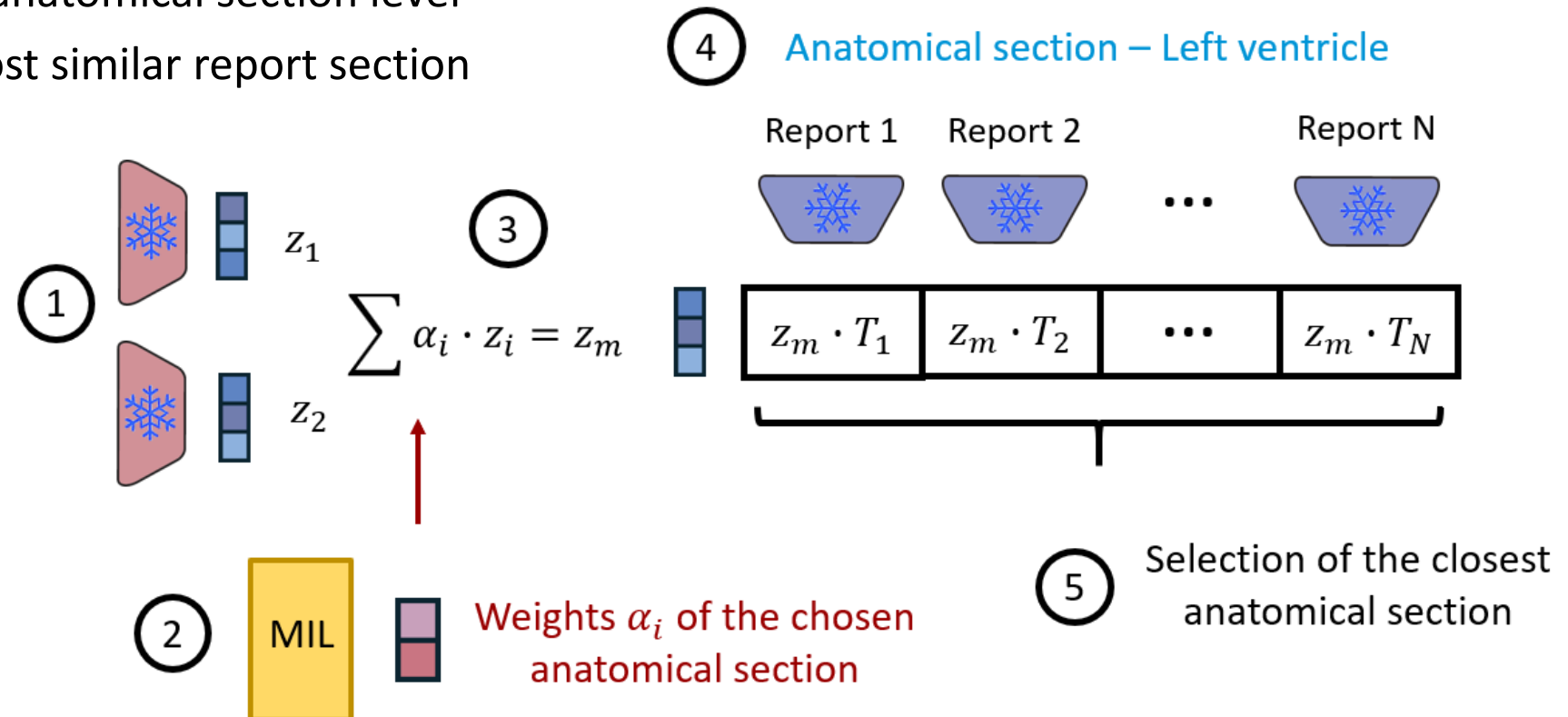
EchoPrime learned attention α_i



Multimodal foundation models - EchoPrime

Cross modal retrieval

- Operate at the anatomical section level
- Retrieve the most similar report section



Multimodal foundation models - EchoPrime



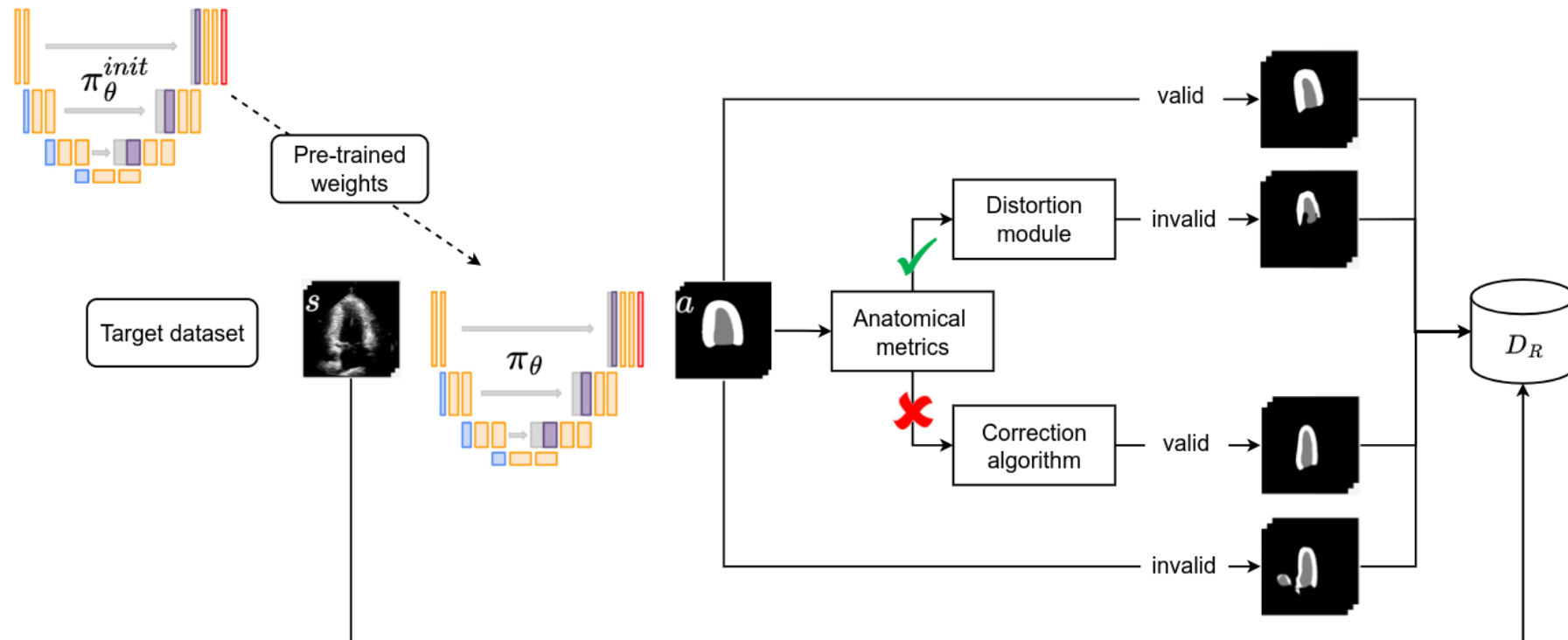
Conclusions

- ✓ Well-suited for semi-automatic annotation
- ✓ Strong generalization across tasks and modalities
- ✓ A powerful starting point for downstream tasks
- ✗ Still less accurate than task-specific methods
- ✗ Efficient multimodal integration remains an open challenge
- ✗ Explainability and robustness remain key challenge
- ✗ Clinical adaptation remains essential

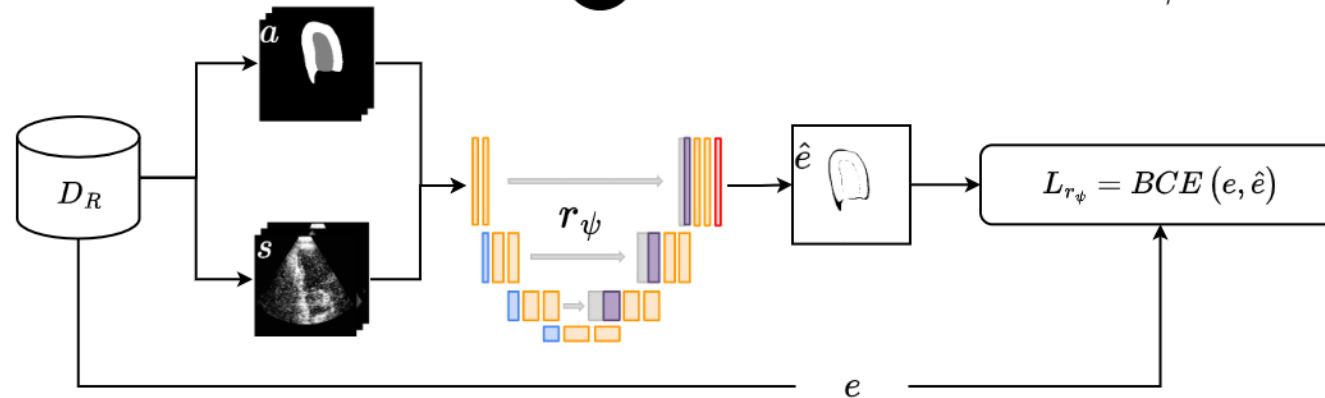
Perspectives - Domain adaptation through reinforcement learning

Optimized model
from the CAMUS dataset

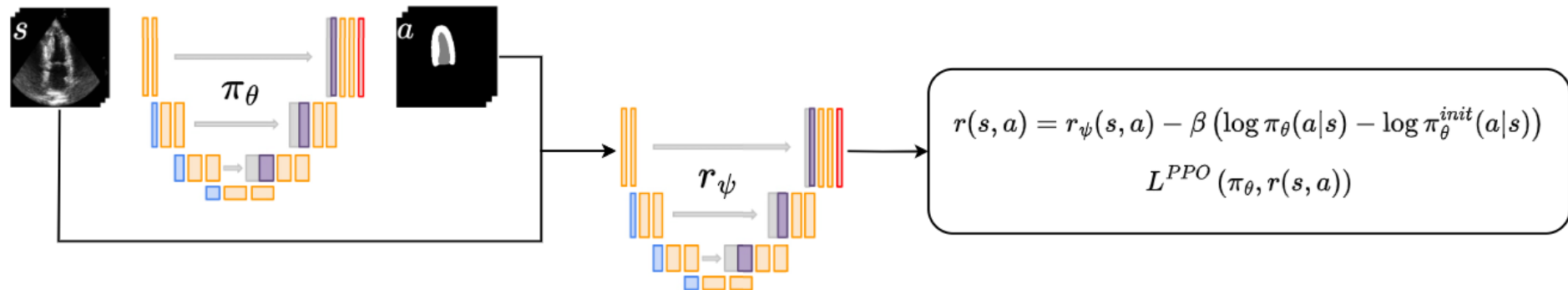
1 Collect actions to create a reward dataset D_R



2 Train a reward network r_ψ



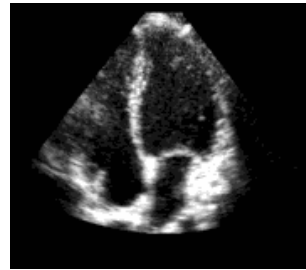
3 Fine-tune the policy network π_θ with PPO algorithm



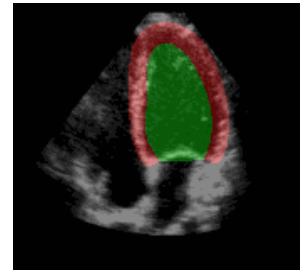
Perspectives - Domain adaptation through reinforcement learning

Successful case

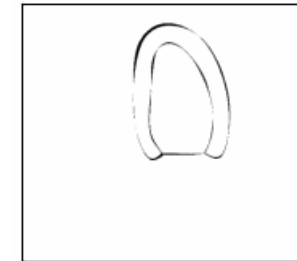
Input image



Segmentation

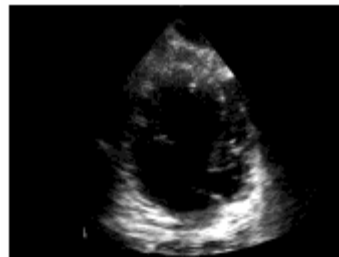


Reward

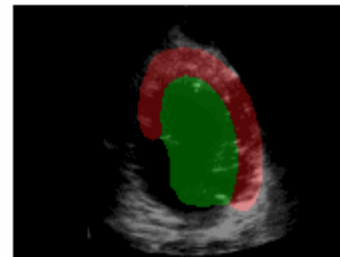


Detected failure
case with correction

Input image



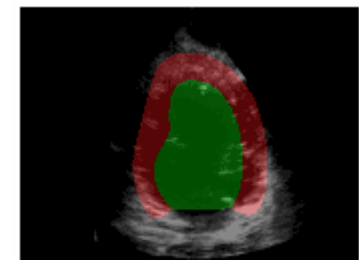
Segmentation



Reward



TTO correction



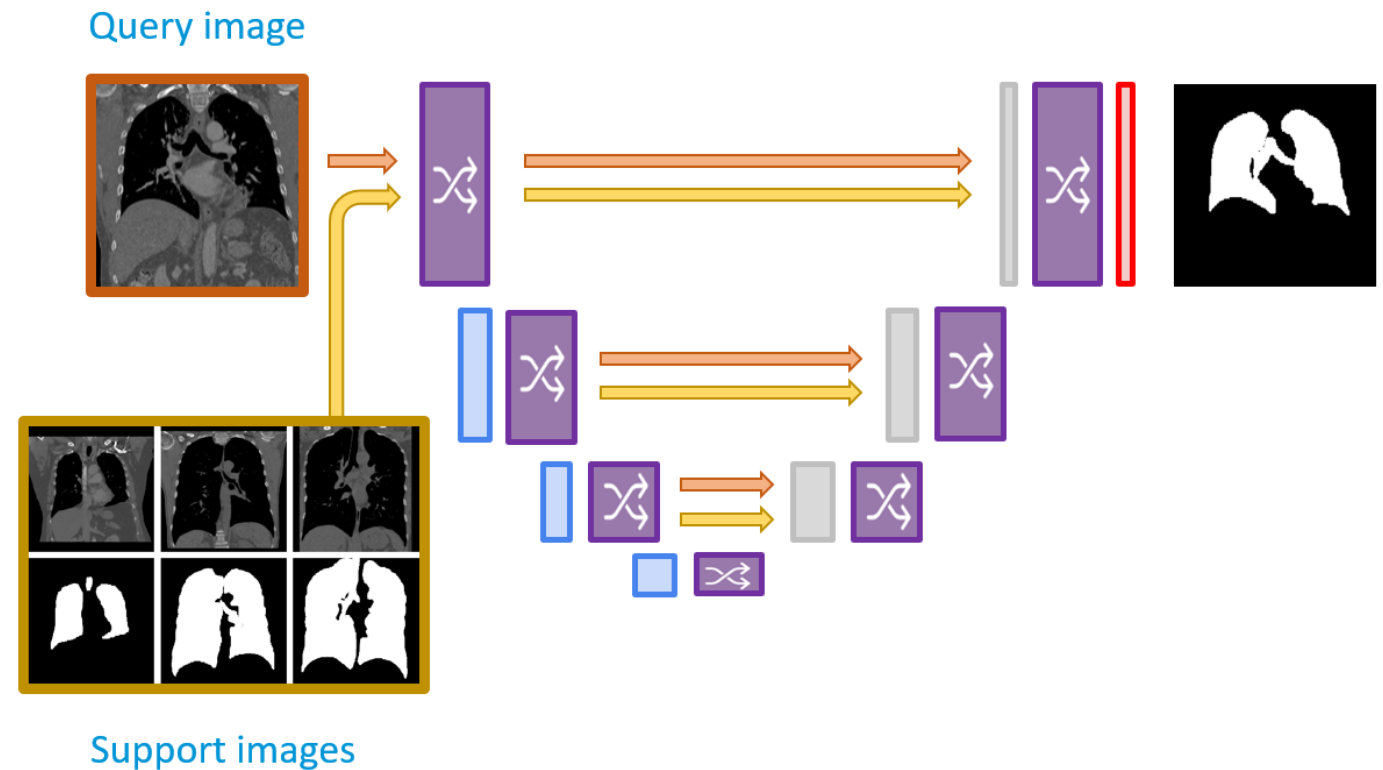
Appendices



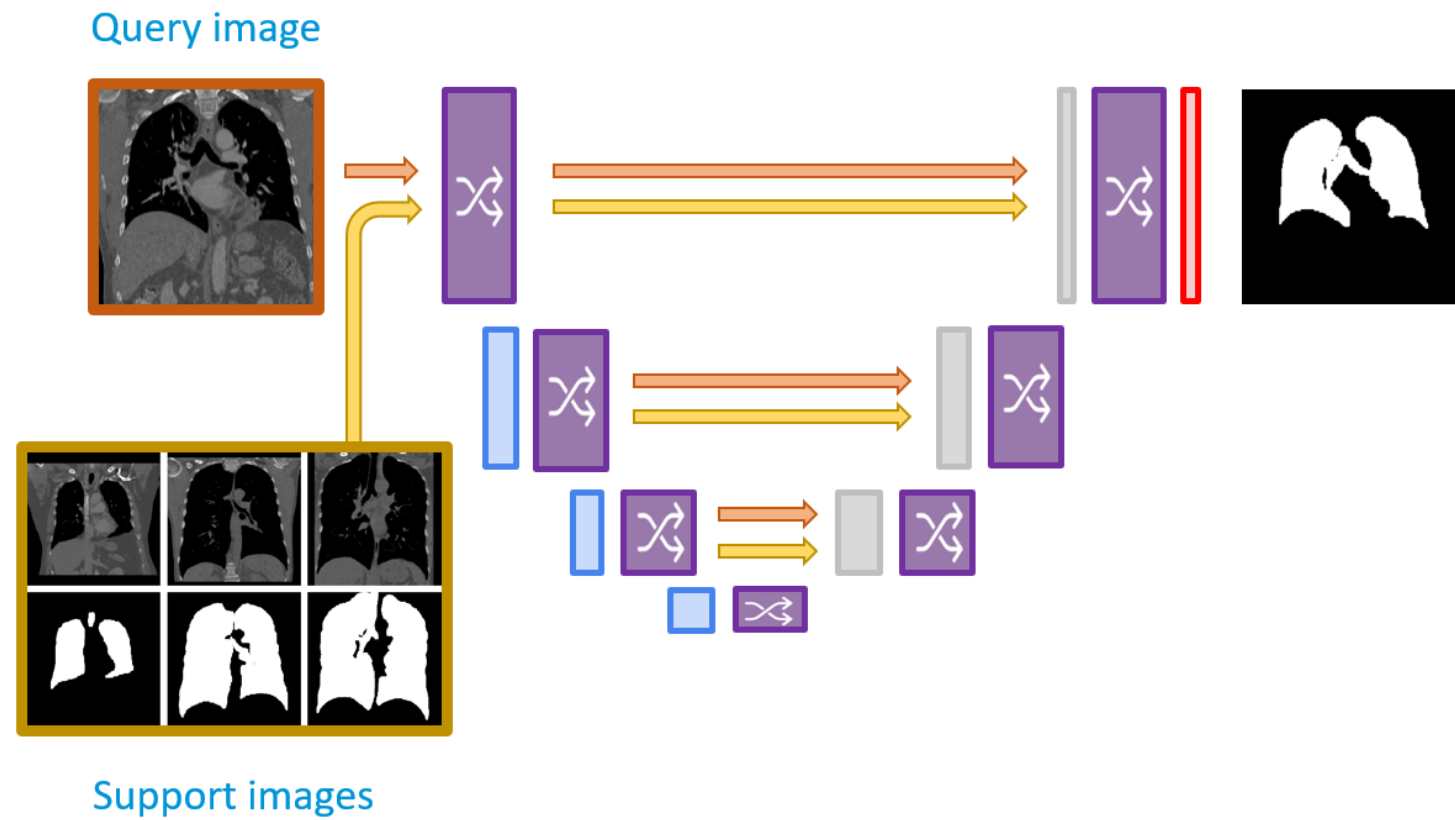
Foundation models - UniverSeg

- ▶ 20 M of parameters
- ▶ Segmentation tasks

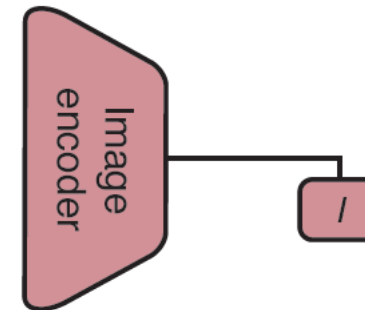
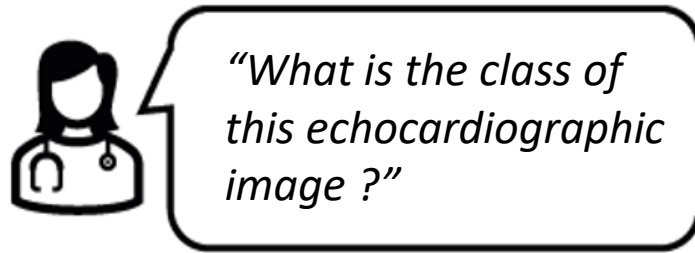
- 22 K images with annotations
- 26 medical domains
- 16 imaging modalities



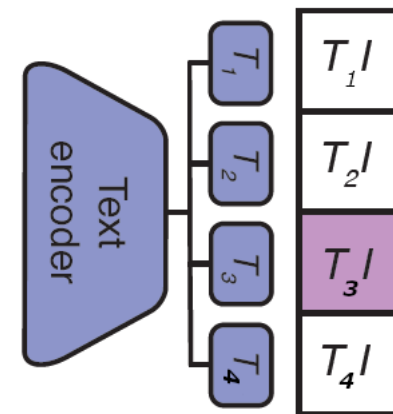
Foundation models - UniverSeg



BiomedCLIP - Image classification



- T1: Hypertensive heart disease
- T2: Cardiac amyloidosis
- T3: Hypertrophic cardiomyopathy
- T4: This is a normal heart



☐ Hypertensive heart disease (10 %)

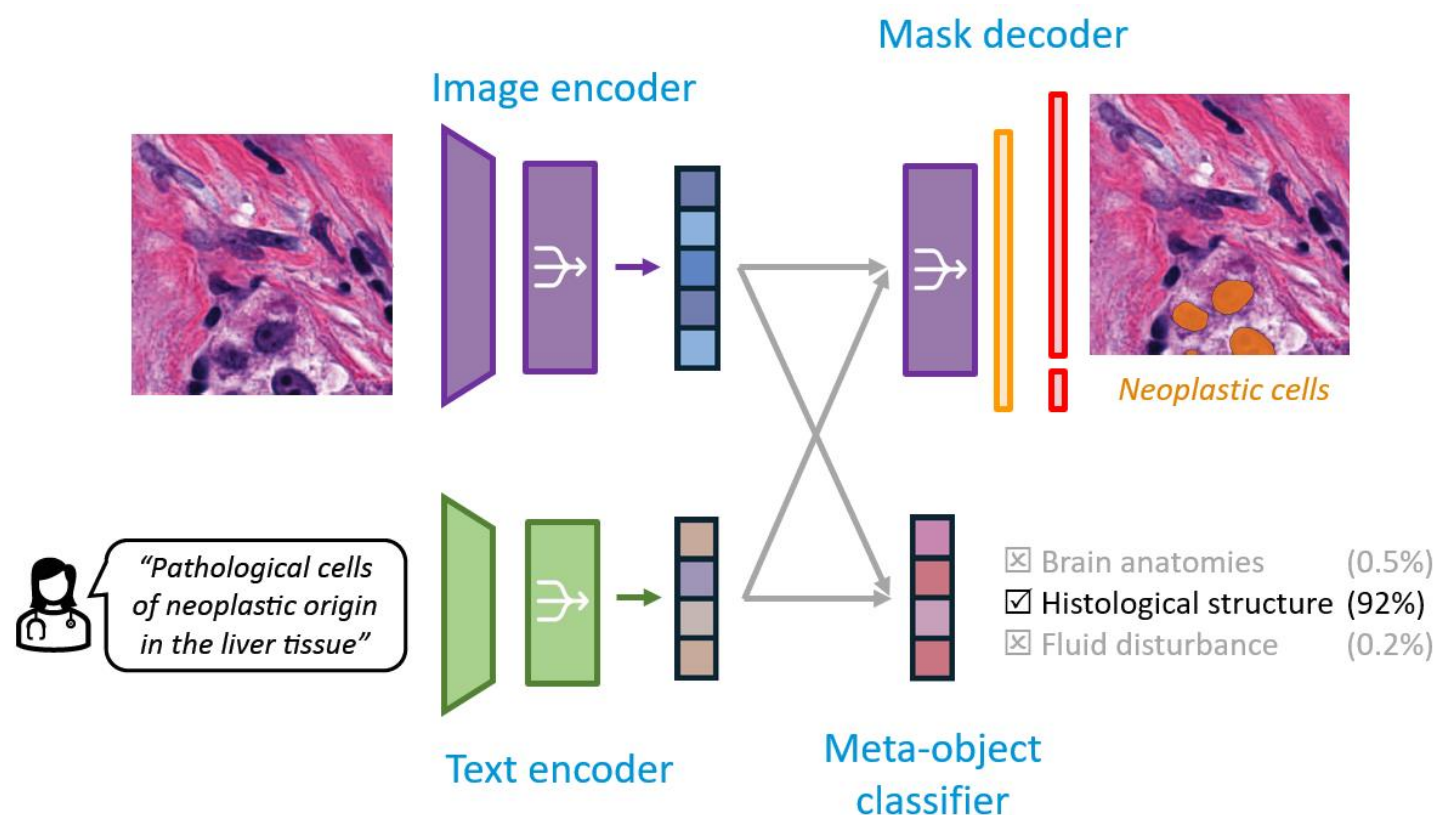
☐ Cardiac amyloidosis (4 %)

☒ Hypertrophic cardiomyopathy (85 %)

☐ This is a normal heart (1 %)

Multimodal foundation models - BiomedParse

- ▶ End to end training
- ▶ 4 losses
 - Dice / focal for mask segmentation
 - Cross-entropy for classification
 - Contrastive for text/image alignment
 - Cross-entropy for meta-object classification



Multimodal foundation models - BiomedParse

► Use of GTP-4 for dataset generation

Harmonization of structure names

- ✓ Biomedical ontology
- ✓ Ex: LV, LV blood pool, left ventricle → left ventricular cavity

Textual descriptor

- ✓ Each mask is associated with a textual descriptor
- ✓ Ex: Liver → Liver in abdominal CT image
- ✓ Ex: LV → left ventricular cavity in echocardiography

Semantic hierarchy

- ✓ glioma → brain tumor → abnormality → pathology
- ✓ Left ventricle → heart chamber → cardiac structure → organ

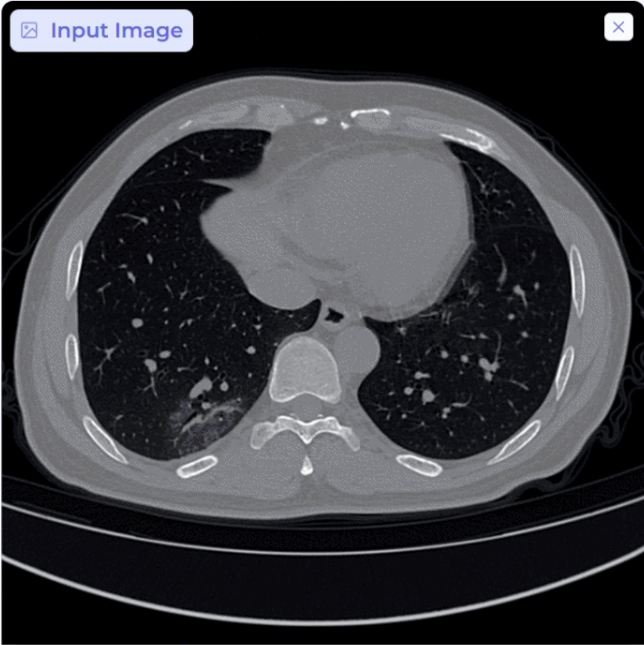
Enhance annotations

- ✓ Ex: Image + mask +
Canonical label: left ventricular myocardium
Description: myocardium of the left ventricle in cardiac MRI
Modality: MRI

BiomedParse - Detection task

From <https://microsoft.github.io/BiomedParse/>

Input Image







Parsing Results



Text prompt

Output text

Go!

Multimodal foundation models - EchoPrime

Anatomical attention module

- Training procedure repeated for each interpretation task within each anatomical section

Interpretation tasks

Anatomical
section

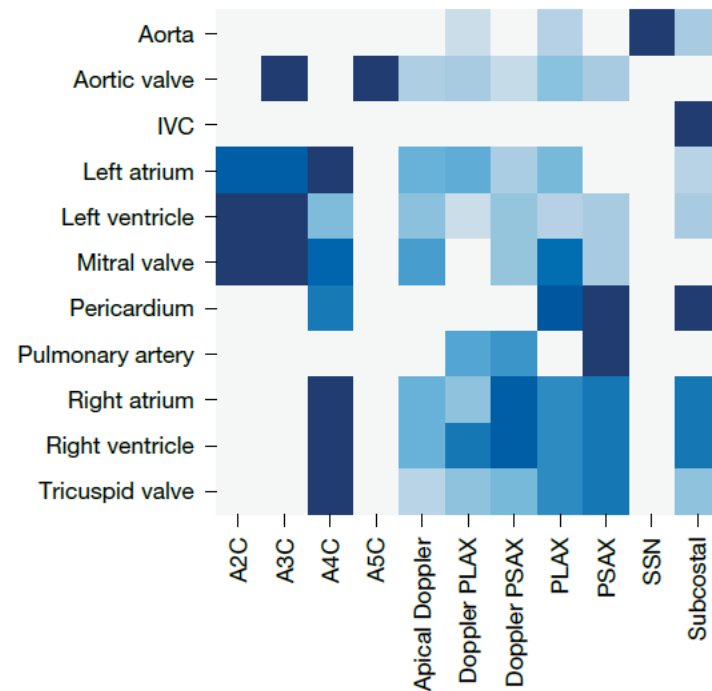
View	LV ejection fraction is 56%	LV size is normal	No LV wall motion abnormality	Left ventricle $\frac{1}{N} \sum \alpha^{(n)}$	
A4C	0.8	0.9	0.7	0.8	$\alpha_1^{(n)}$
Apical Doppler	0.2	0.1	0.3	0.2	$\alpha_2^{(n)}$

Aorta	0.2	0.1	0.2	0.4
Left ventricle	0.7	0.9	0.8	0.5
Right atrium	0.1	0.8	0.5	0.8
	A2C	A4C	Apical Doppler	Doppler PLAX

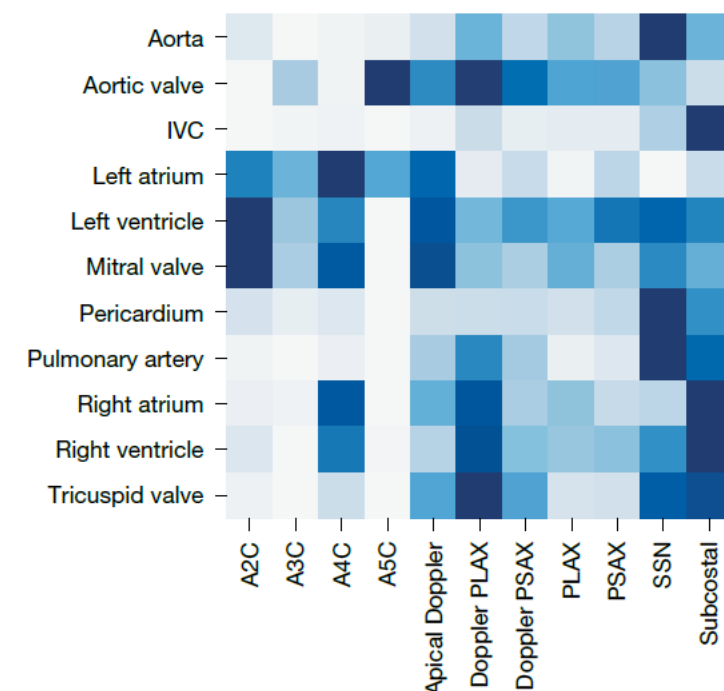
Multimodal foundation models - EchoPrime

Anatomical attention module

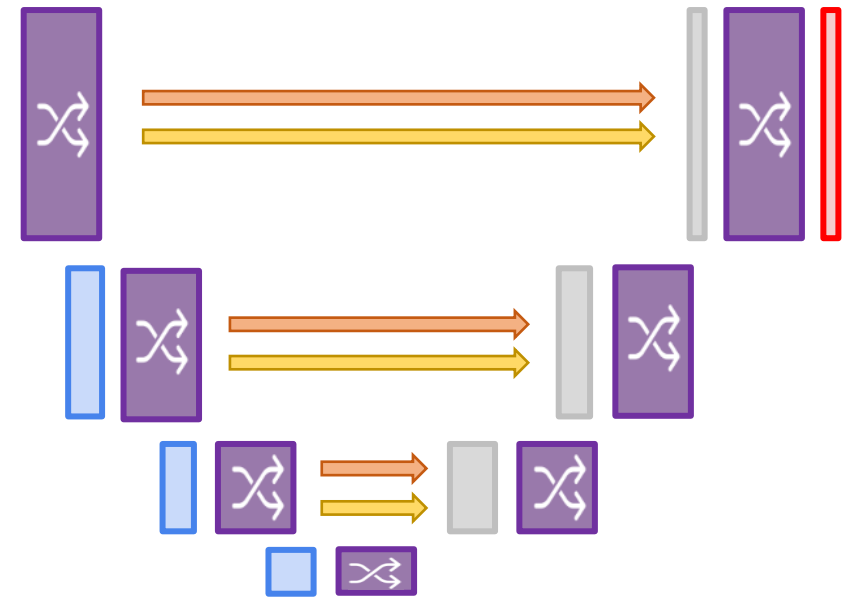
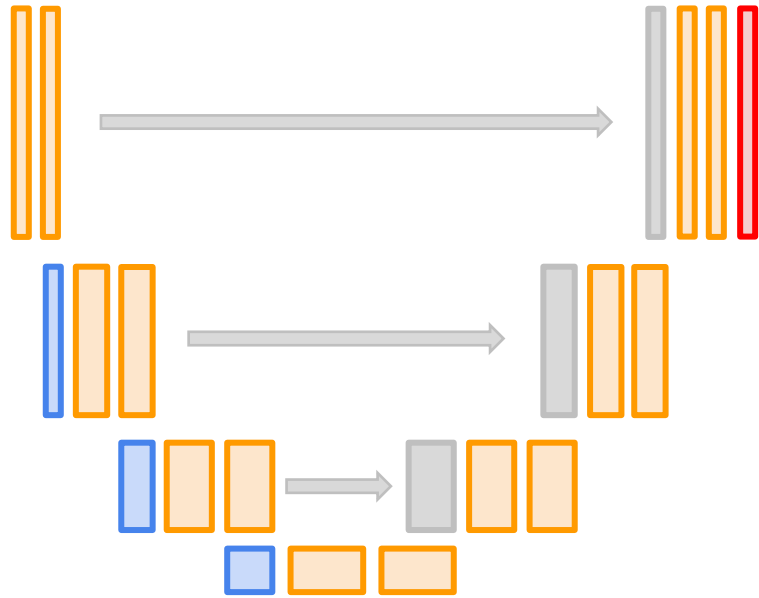
- Prioritization of echocardiographic views for different anatomical structures

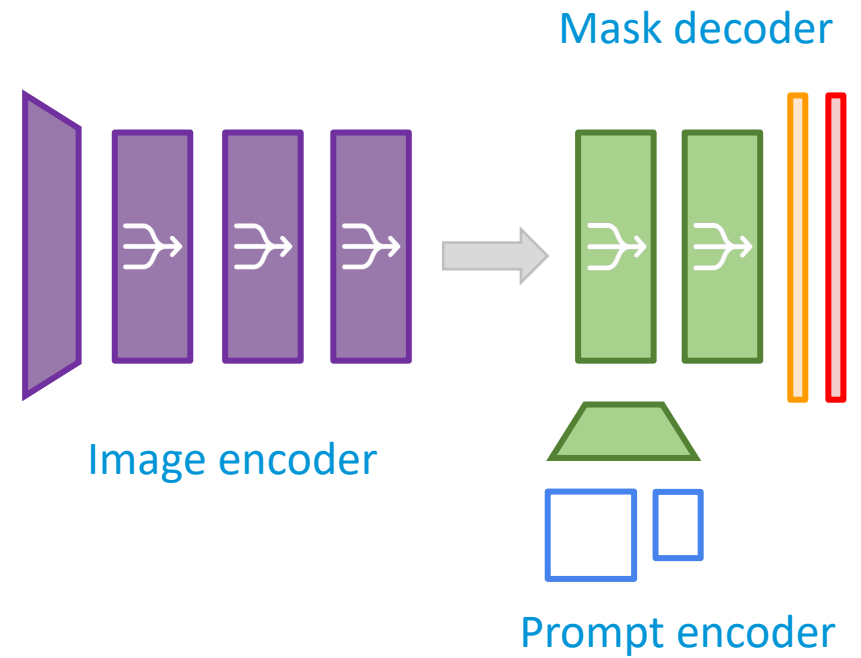
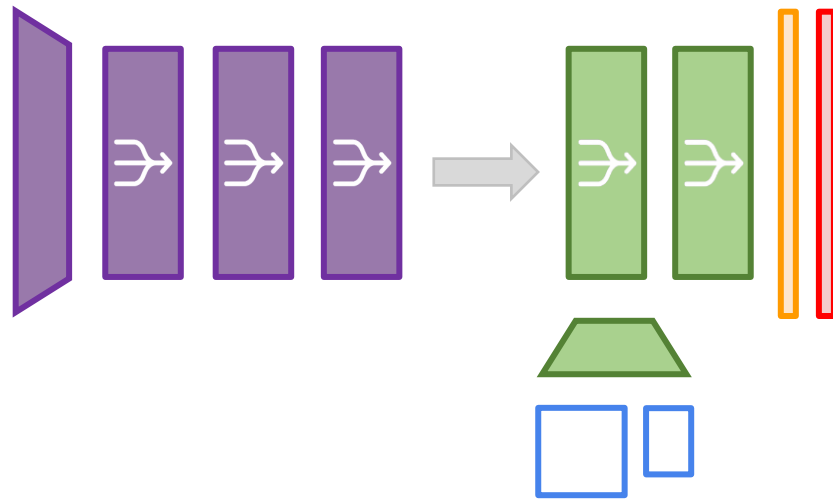


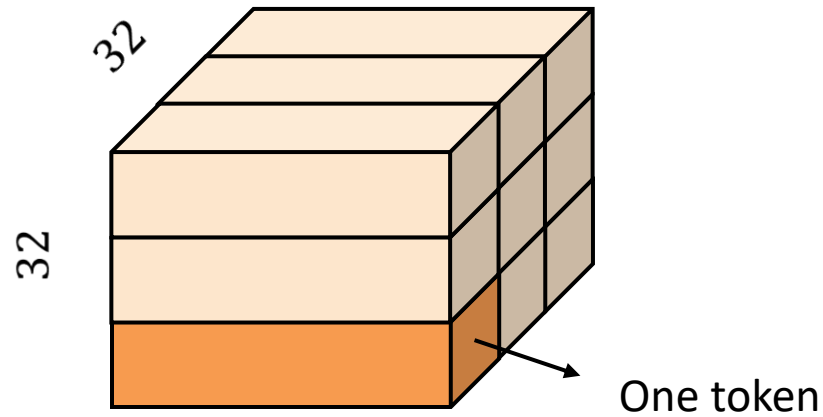
Cardiologist reference



EchoPrime learned attention



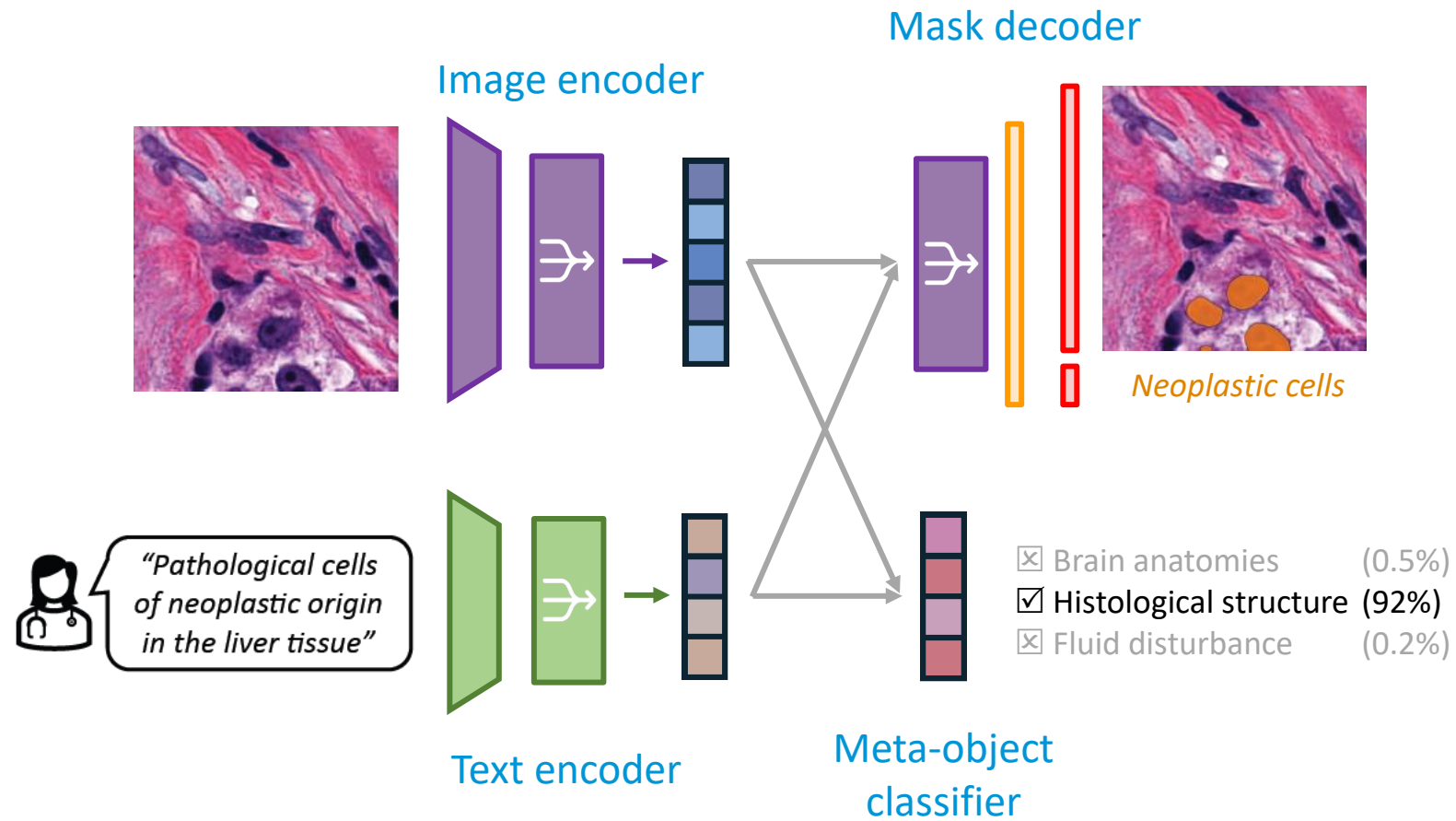


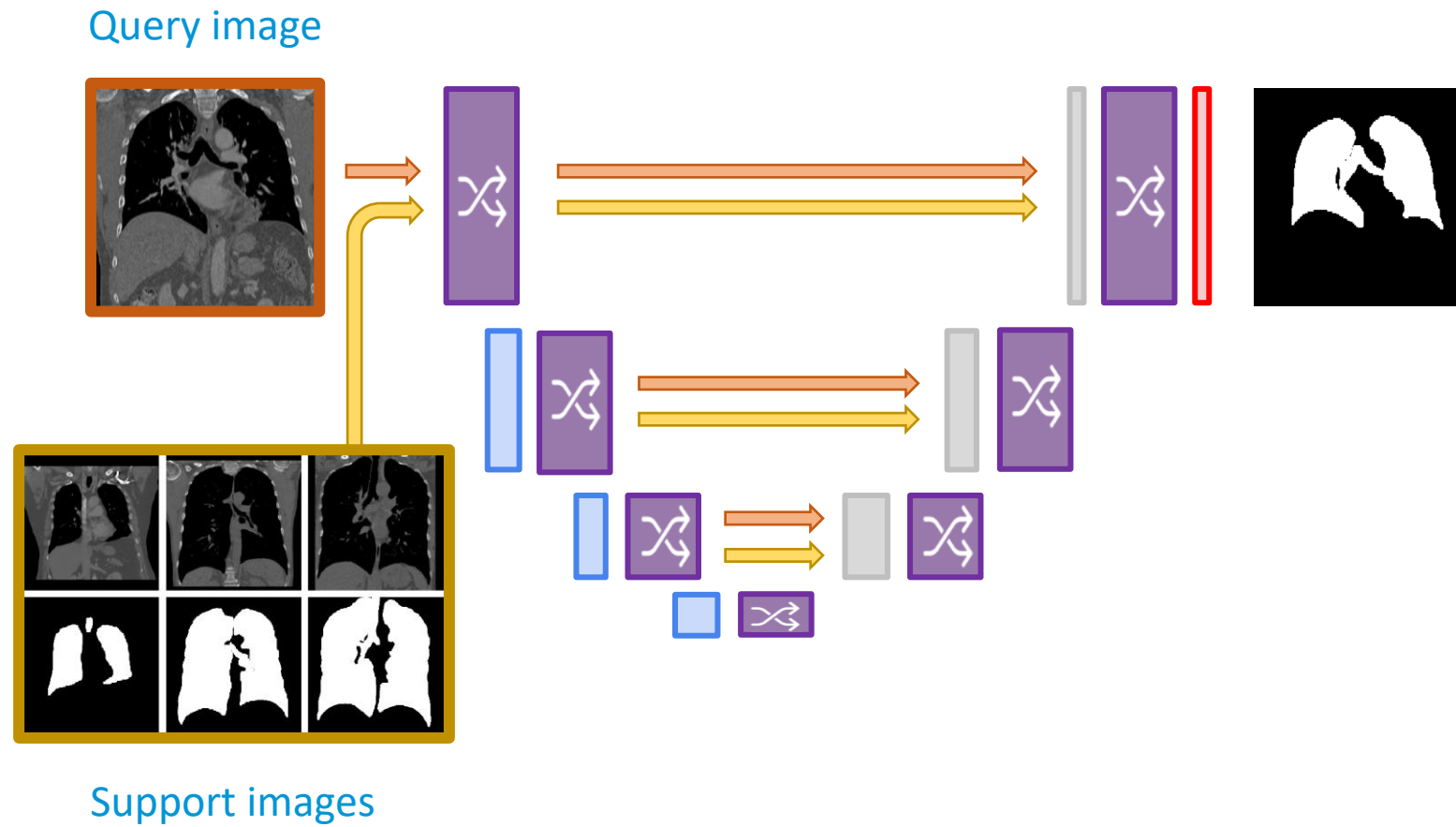


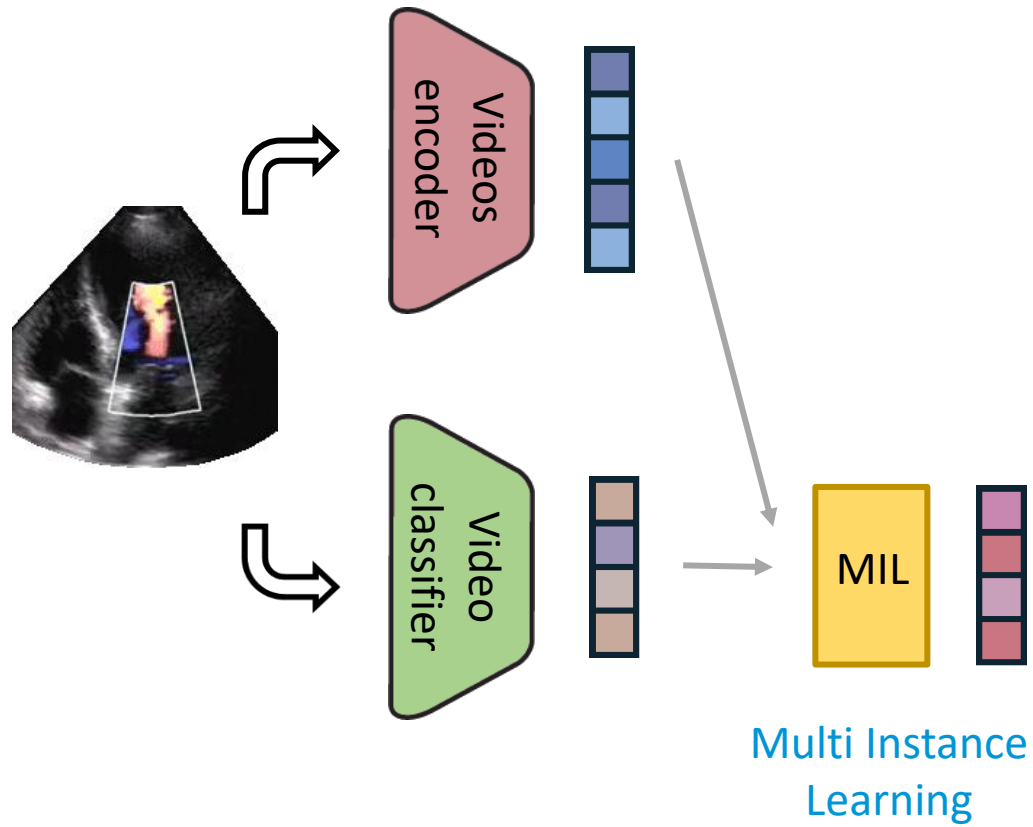
Feature maps

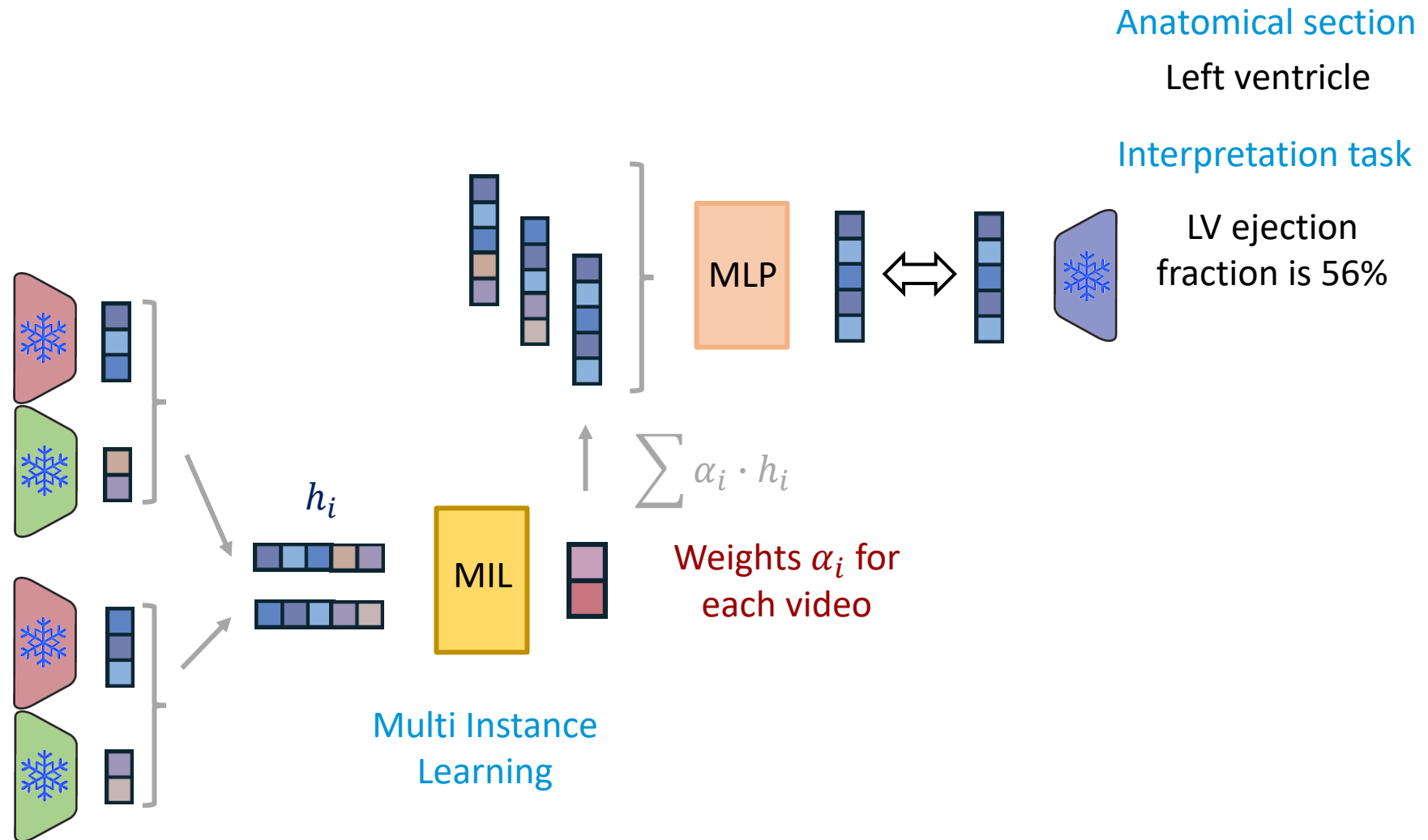


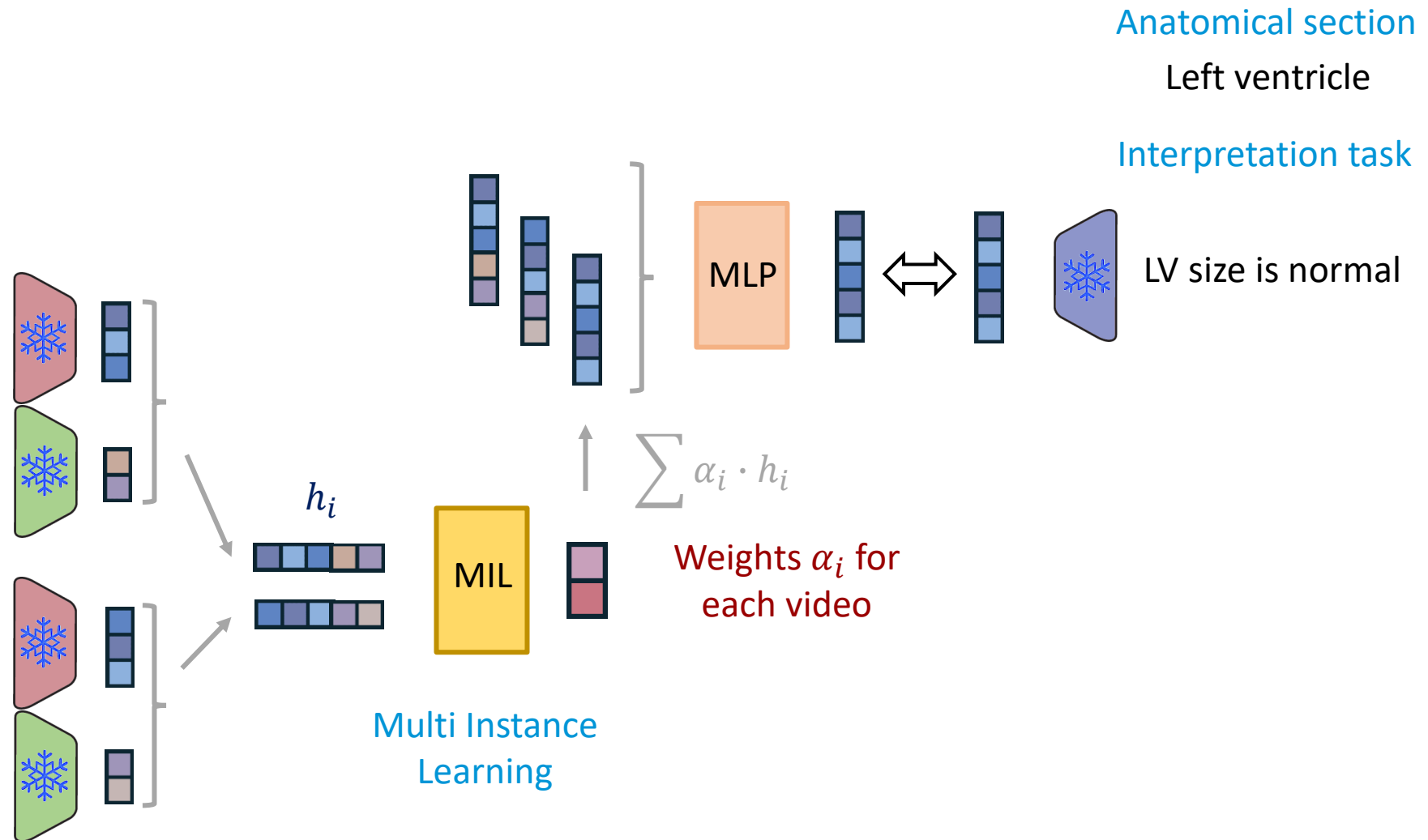
*"Pathological cells
of neoplastic origin
in the liver tissue"*

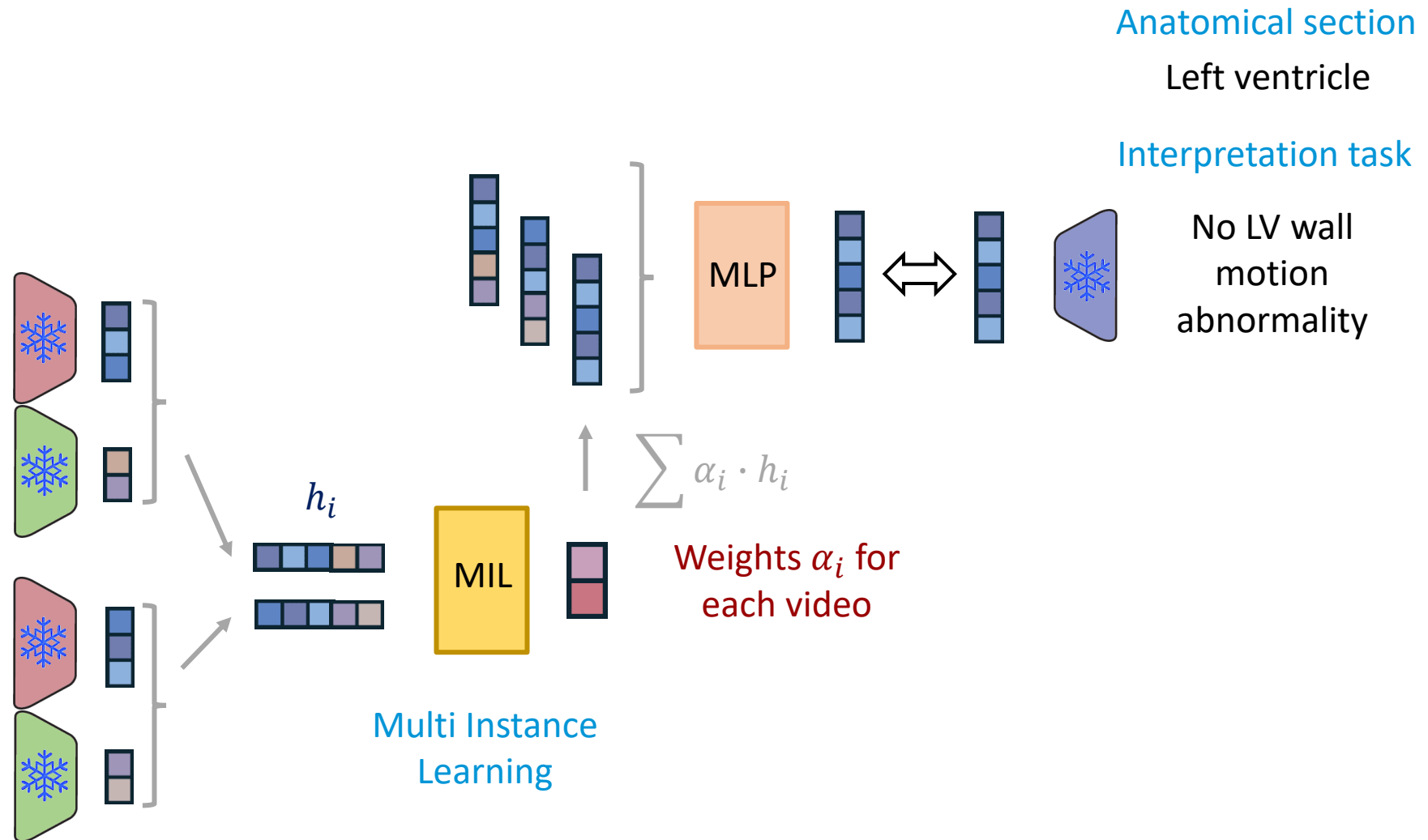


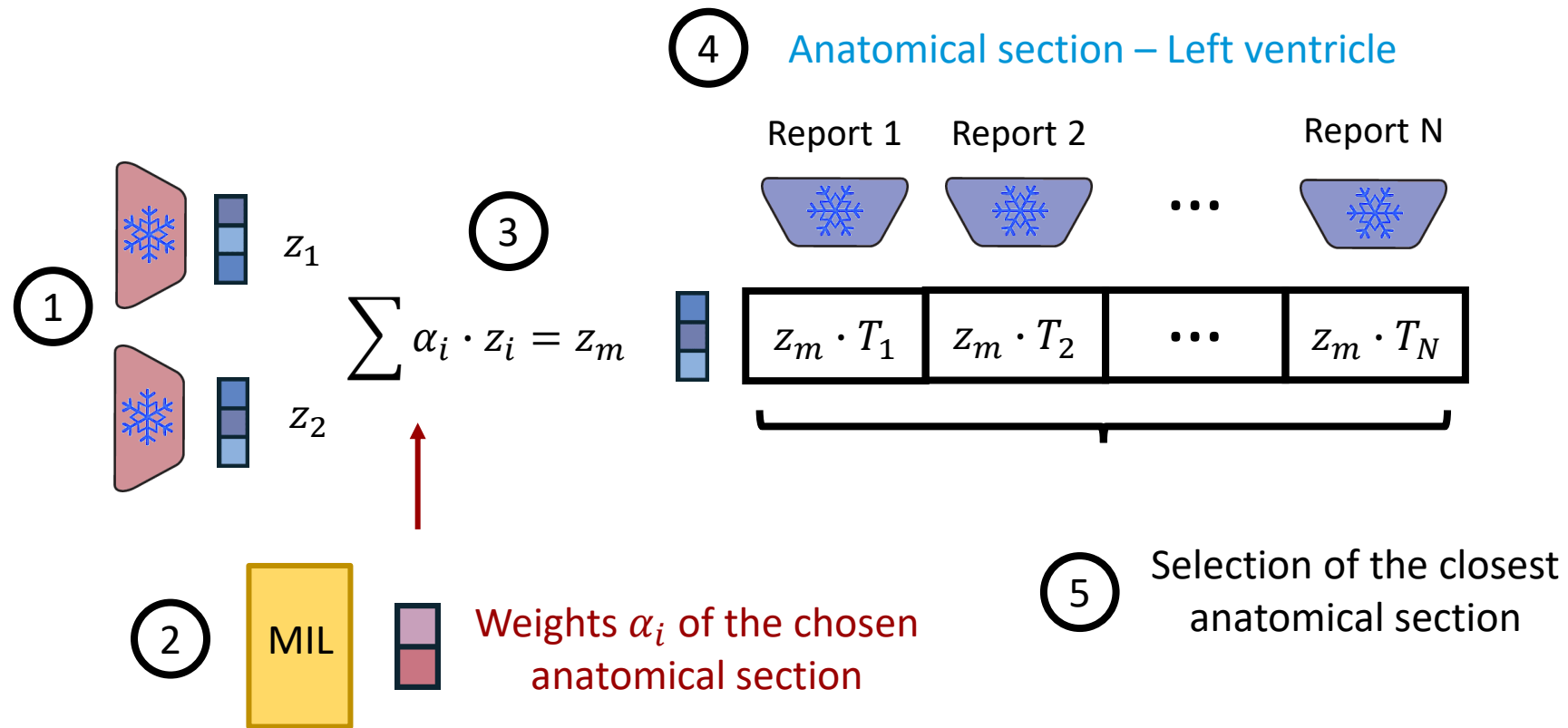










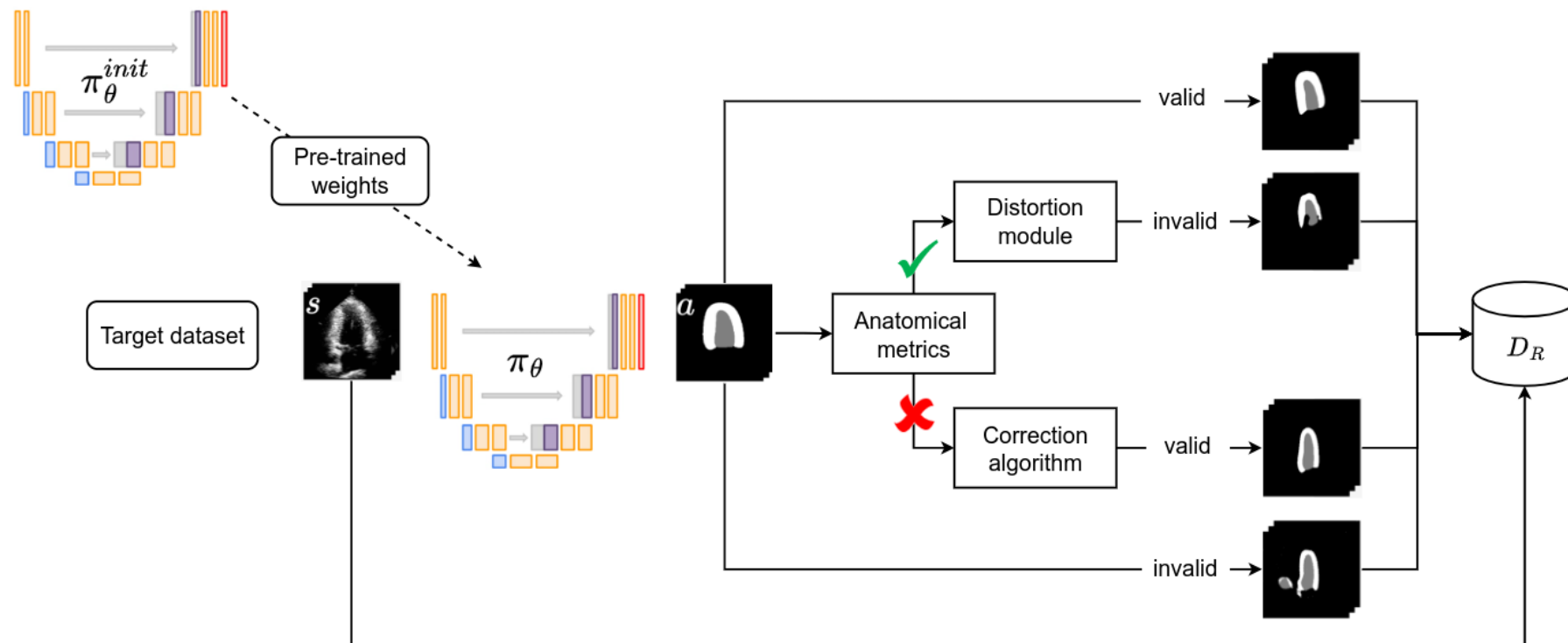


Inspired from reinforcement learning

- ✓ Update nnU-Net weights to fit with the target dataset

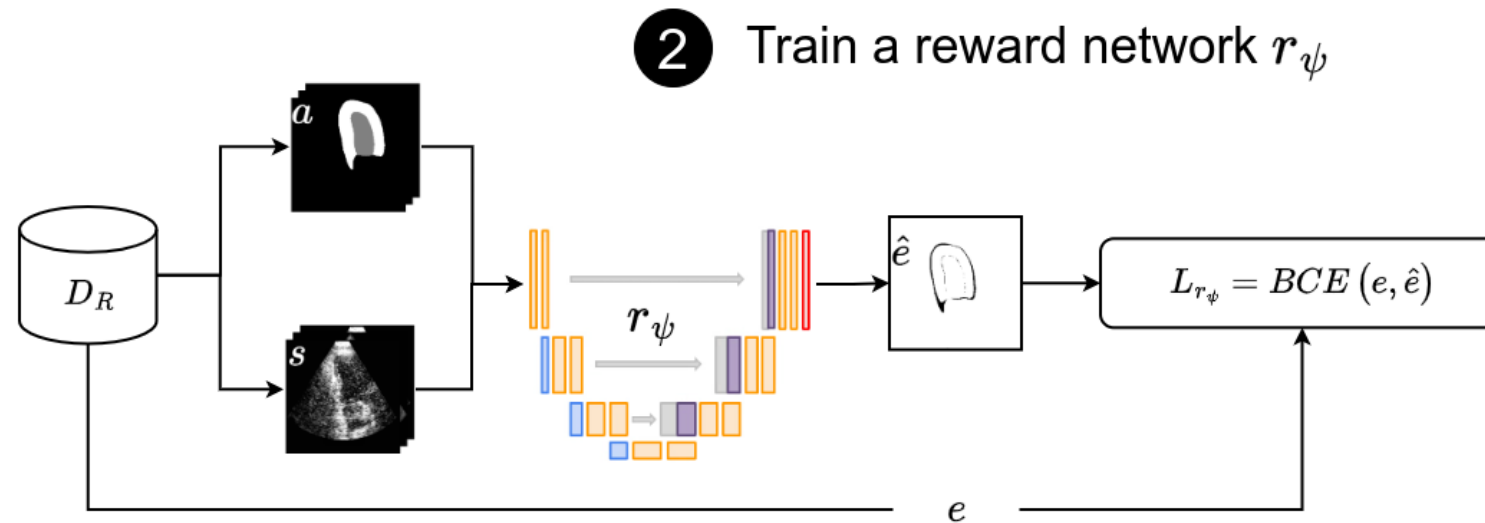
Optimized model
from the CAMUS dataset

1 Collect actions to create a reward dataset D_R



Inspired from reinforcement learning

- ✓ Update nnU-Net weights to fit with the target dataset



Inspired from reinforcement learning

- ✓ Update nnU-Net weights to fit with the target dataset

3 Fine-tune the policy network π_θ with PPO algorithm

